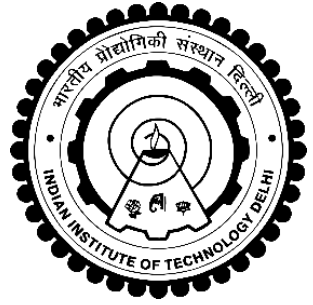




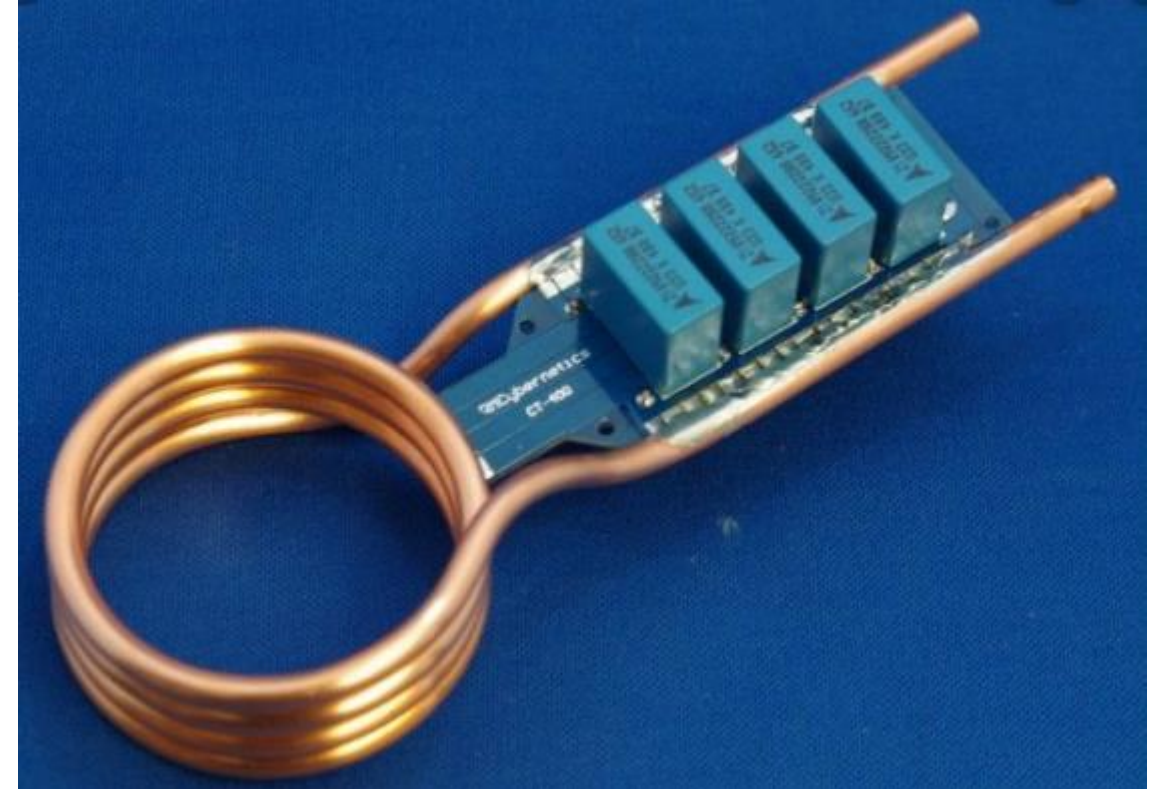
# **ELL 100 – Introduction to Electrical Engineering**

## **LECTURE 16: RESONANCE IN AC CIRCUITS**

# Outline

- ❑ Resonance in Series  $RLC$  Circuit
- ❑ Quality Factor ( $Q$ )
- ❑ Bandwidth and Half-Power Frequencies
  
- ❑ Resonance in Parallel  $RLC$  Circuit
- ❑ Parallel LC Circuit (Tank Circuit)
- ❑ Dynamic Impedance (or) Dynamic Resistance

# PHENOMENON OF RESONANCE IN VARIOUS DOMAINS



Induction Heater

# PHENOMENON OF RESONANCE IN VARIOUS DOMAINS



RF Amplifiers

# PHENOMENON OF RESONANCE IN VARIOUS DOMAINS

TV Receivers





# PHENOMENON OF RESONANCE IN VARIOUS DOMAINS



Radio

# PHENOMENON OF RESONANCE IN VARIOUS DOMAINS



Guitar

# PHENOMENON OF RESONANCE IN VARIOUS DOMAINS



Pendulum



# PHENOMENON OF RESONANCE IN VARIOUS DOMAINS



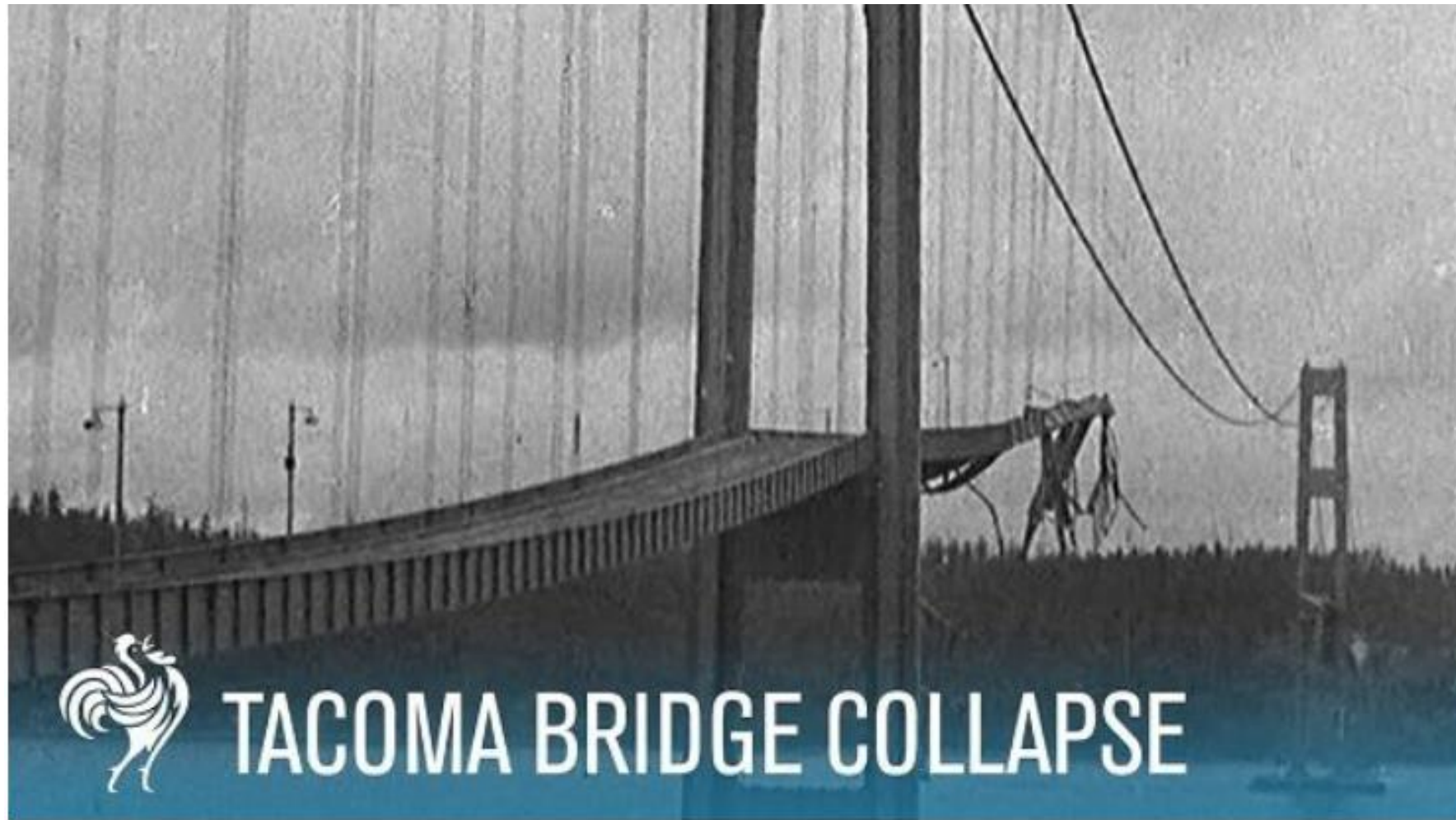
Swing

# PHENOMENON OF RESONANCE IN VARIOUS DOMAINS



Singer Breaking the Glass

# PHENOMENON OF RESONANCE IN VARIOUS DOMAINS



## Bridge Collapse

# PHENOMENON OF RESONANCE IN VARIOUS DOMAINS



Microwave Oven

# WHY STUDY RESONANCE?

- Resonance is the frequency response of a circuit or network when it is operating at its natural frequency called “ Resonance Frequency”.
- For many applications, the supply (defined by its voltage and frequency) is constant. e.g. The supply to residential homes is 230 V, 50 Hz.
- However, many communication systems involve circuits in which the supply voltage operates with a varying frequency.
- To understand communication systems, one requires a knowledge of how circuits are affected by a variation of the frequency. Examples of such communication systems are,  
Radio, television, telephones, and machine control systems.

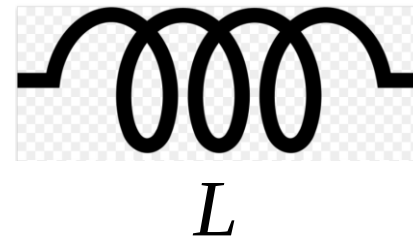


# WHEN RESONANCE OCCURS? AND WHAT IT RESULTS?

- Resonance occurs in any circuit that has energy storage elements, **at least one inductor and one capacitor**.
- Under resonance, **the total impedance is equal to the resistance only** and **maximum power is drawn** from the supply by the circuit.
- Under resonance, the total **supply voltage and supply current are in phase**. So, the **power factor (PF) becomes unity**.
- At resonance,  $L$  and  $C$  elements exchange energy freely as a function of time, which results in sinusoidal oscillations either across  $L$  or  $C$ .

## TYPES OF RESONANCE

- Series resonance.
- Parallel resonance.





# APPLICATIONS OF RESONANCE

- Resonant circuits (series or parallel) are used in many applications such as selecting the desired stations in radio and TV receivers.
- Most common applications of resonance are based on the frequency dependent response. (“tuning” into a particular frequency/channel)
- A series resonant circuit is used as voltage amplifier.
- A parallel resonant circuit is used as current amplifier.
- A resonant circuit is also used as a filter.

# RESONANCE IN SERIES *RLC* CIRCUIT

**Resonance** is a condition in an *RLC* circuit in which the **capacitive and inductive reactances are equal** in magnitude, thereby resulting in a purely resistive impedance.

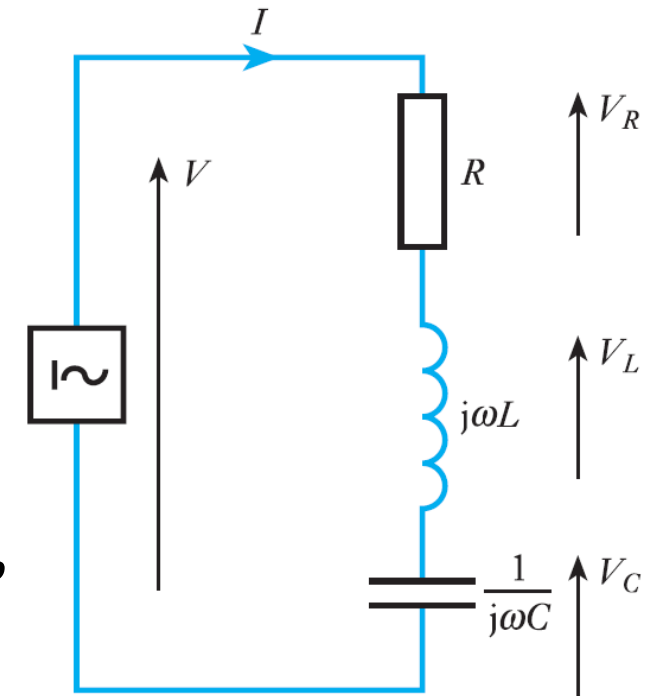
The input impedance is as follows,

$$Z = R + j\omega L + \frac{1}{j\omega C} = R + j\left(\omega L - \frac{1}{\omega C}\right)$$

At resonance, the net reactance becomes zero. Therefore,

$$\omega_r L = \frac{1}{\omega_r C} \Rightarrow \omega_r = \frac{1}{\sqrt{LC}} \text{ rad/s}; f_r = \frac{1}{2\pi\sqrt{LC}} \text{ Hz}$$

where  $\omega_r$  and  $f_r$  represent resonant frequency in rad/s and in Hz, respectively



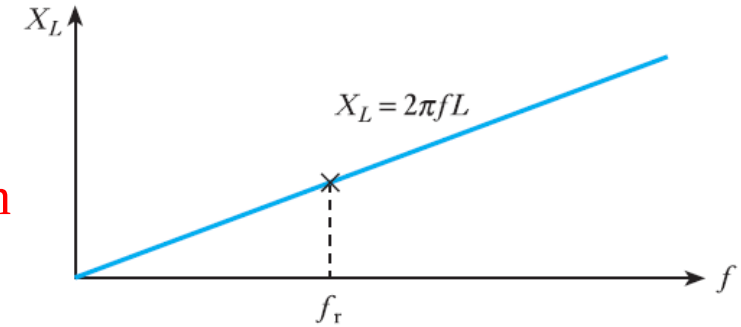
Series resonant *RLC* circuit

# REACTANCE ( $X_L, X_C$ ) VS FREQUENCY PLOTS

The value of the reactance  $X$  of the circuit is,

$$X = \omega L - \frac{1}{\omega C} \rightarrow \text{Depends on frequency}$$

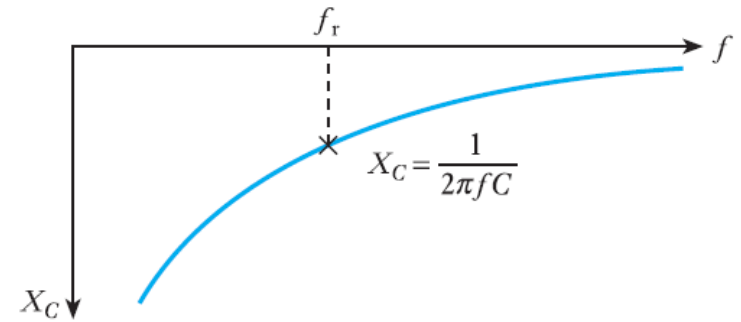
Variation of  
inductive  
reactance with  
frequency



The inductive reactance:

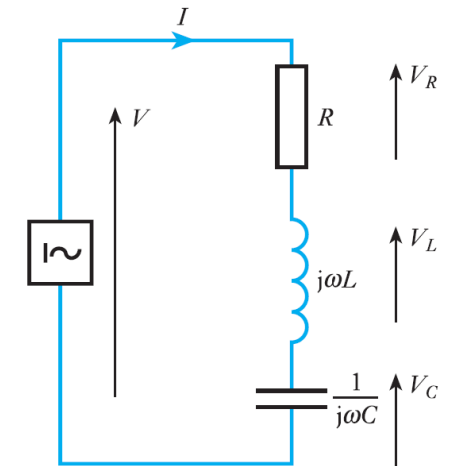
$$|X_L| = \omega L = 2\pi fL \rightarrow \text{Increases linearly with frequency}$$

Variation of  
Capacitive  
reactance with  
frequency



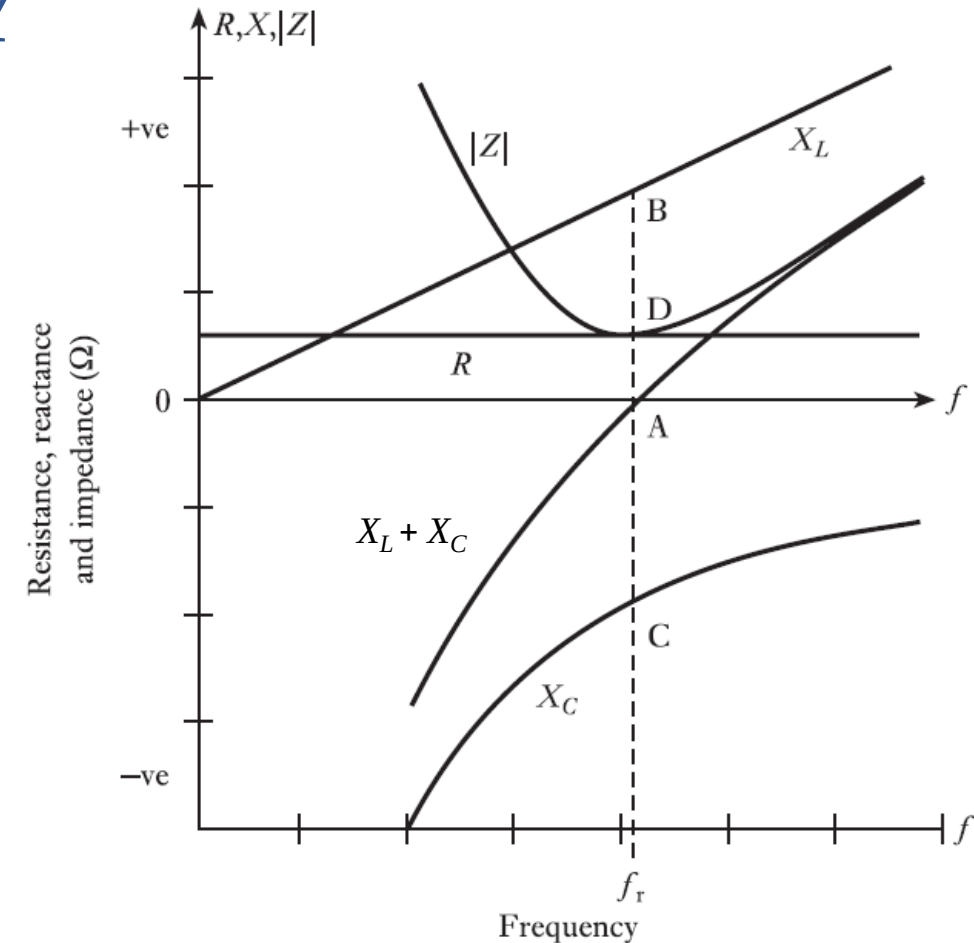
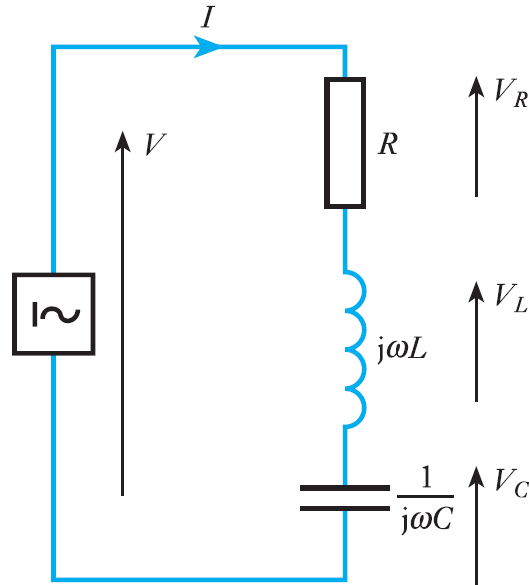
The capacitive reactance:

$$|X_C| = \frac{1}{\omega C} = \frac{1}{2\pi fC} \rightarrow \text{Decreases with frequency and it is largest at low frequencies}$$



# VARIATION OF REACTANCE AND IMPEDANCE WITH FREQUENCY

- At resonant frequency  $f_r$ ,  $|Z| = R$ , the power factor is unity (**purely resistive**).
- Below  $f_r$ ,  $|X_L| < |X_C|$ , so the circuit is more **capacitive** and the power factor is leading.
- Above  $f_r$ ,  $|X_L| > |X_C|$ , so the circuit is more **inductive** and the power factor is lagging.

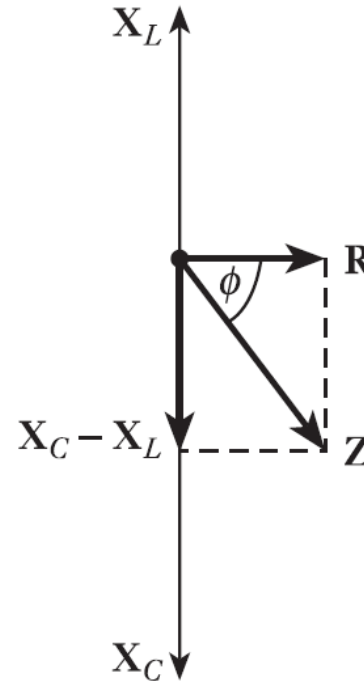
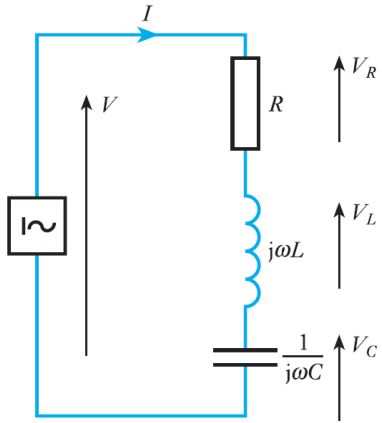


Variation of resistance, reactance and impedance with frequency

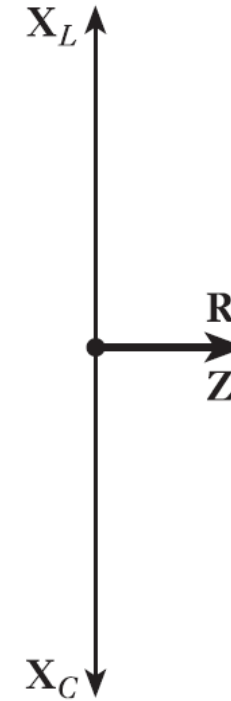
# IMPEDANCE PHASOR DIAGRAMS

The phase of the circuit impedance is given by

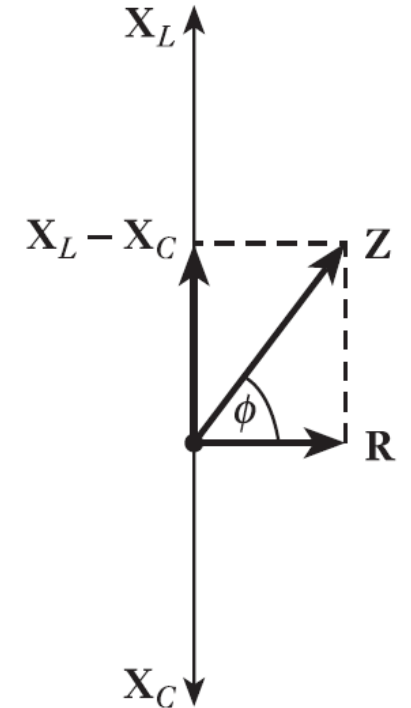
$$\phi = \tan^{-1} \frac{(X_L - X_C)}{R}$$



Below  $f_r$ ,  $X_C > X_L$



At  $f_r$ ,  $X_C = X_L$ ,  $Z = R$



Above  $f_r$ ,  $X_L > X_C$

- Below  $f_r$ ,  $X_L < X_C$ ,  $\phi$  is negative, the circuit is capacitive.
- At resonance ( $f_r$ ),  $X_L = X_C$ ,  $\phi$  is zero, the circuit is purely resistive.
- Above  $f_r$ ,  $X_L > X_C$ ,  $\phi$  is positive, the circuit is inductive.

# THE CURRENT IN A SERIES *RLC* CIRCUIT

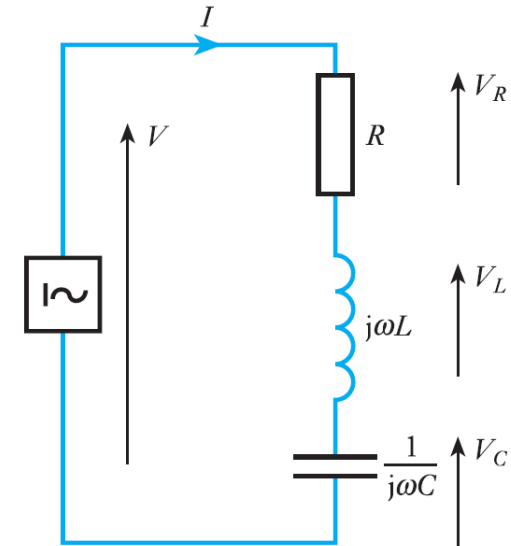
The circuit current is given by

$$I = \frac{V}{Z \angle \phi} = \frac{V \angle -\phi}{Z}$$

$$I = \frac{V}{\sqrt{\left\{ R^2 + \left( \omega L - \frac{1}{\omega C} \right)^2 \right\}}} \angle -\tan^{-1} \left( \frac{\omega L - \frac{1}{\omega C}}{R} \right)$$

The current is maximum when  $\omega L = 1/(\omega C)$ , when the circuit is resistive ( $\phi = 0$ ). Therefore,

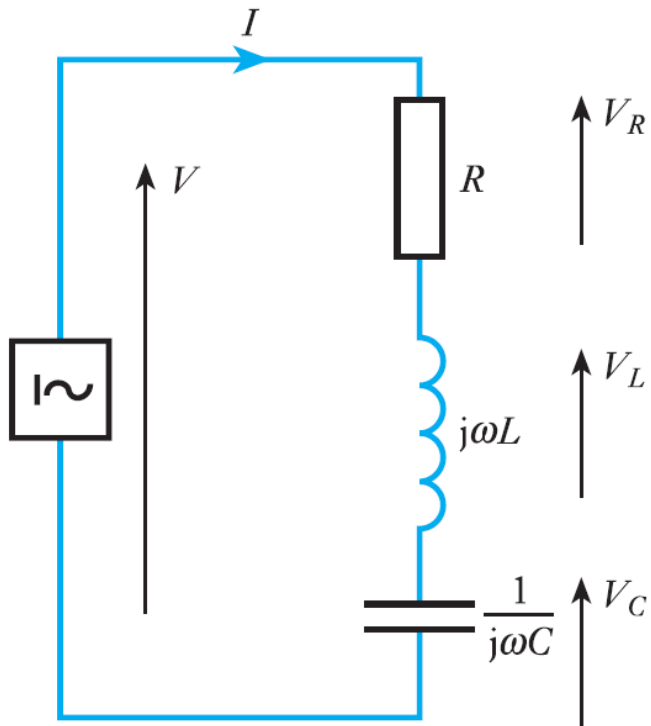
$$I_m = \frac{V}{R}$$



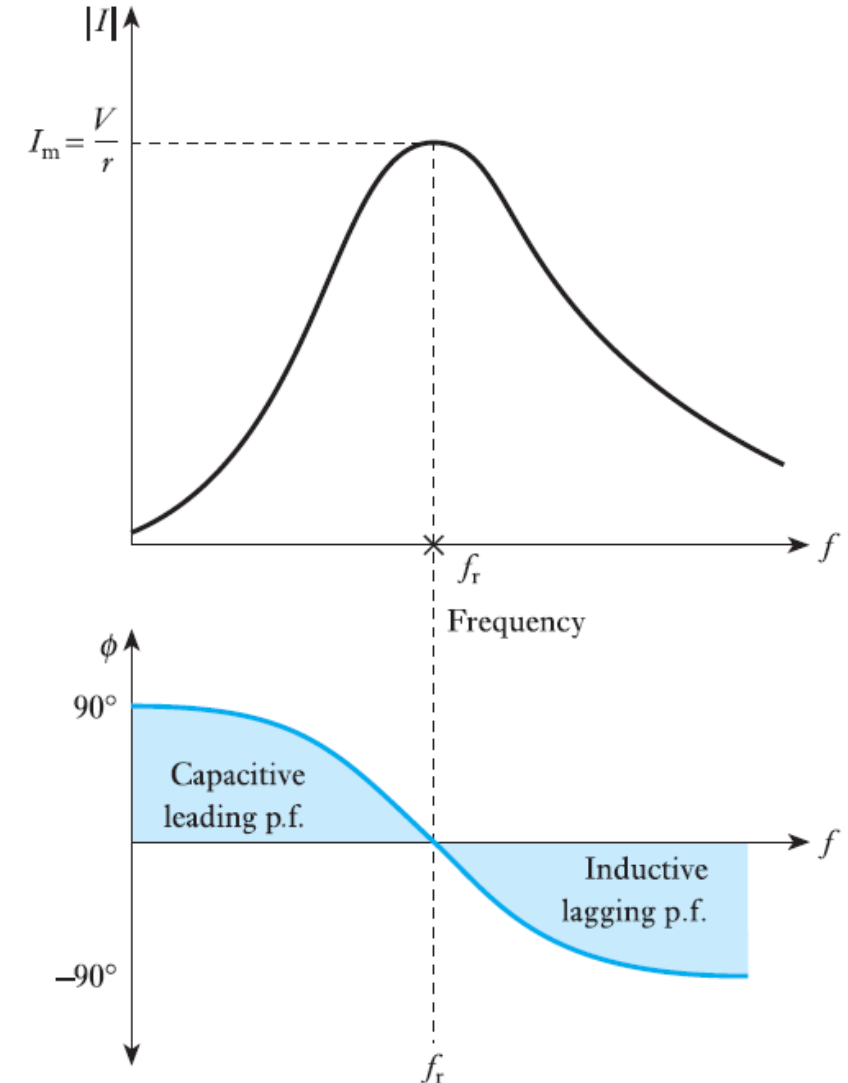


# VARIATION OF MAGNITUDE AND PHASE OF CURRENT WITH FREQUENCY

- The current is maximum at resonant frequency ( $f_r$ ).



Variation of magnitude  $|I|$  and phase  $\phi$  of current with frequency in a series RLC circuit



# QUALITY FACTOR (Q)

- The “sharpness” of the resonance in a resonant circuit is measured quantitatively by the quality factor  $Q$ .
- The quality factor relates the maximum or peak energy stored to the energy dissipated in the circuit per cycle of oscillation:

$$Q = 2\pi \left( \frac{\text{Peak energy stored in the circuit}}{\text{Energy dissipated by the circuit in one period at resonance}} \right)$$

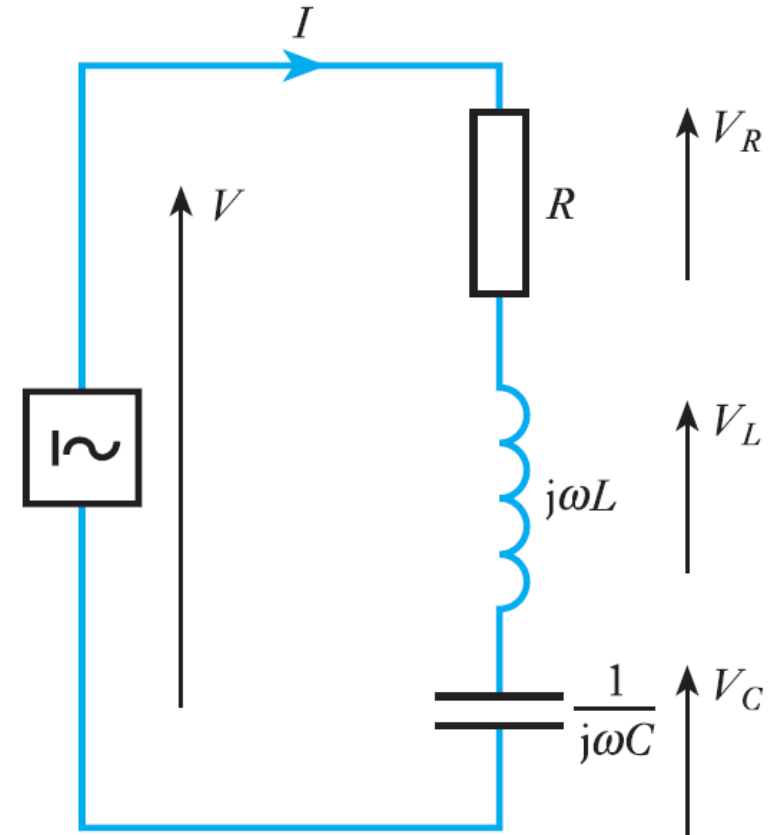
- It is also regarded as a measure of the energy storage property of a circuit in relation to its energy dissipation property.

# QUALITY FACTOR ( $Q$ )

- In the series  $RLC$  circuit, the quality factor ( $Q$ ) is,

$$Q = 2\pi \left( \frac{\frac{1}{2}LI^2}{\frac{1}{2}I^2R\left(\frac{1}{f_r}\right)} \right) = \frac{2\pi f_r L}{R}$$

$$Q = \frac{\omega_r L}{R} = \frac{1}{\omega_r CR} = \frac{1}{R} \sqrt{\frac{L}{C}}$$



# QUALITY FACTOR (Q)

- The  $Q$  factor is also defined as the ratio of the reactive power, of either the capacitor or the inductor to the average power of the resistor at resonance:

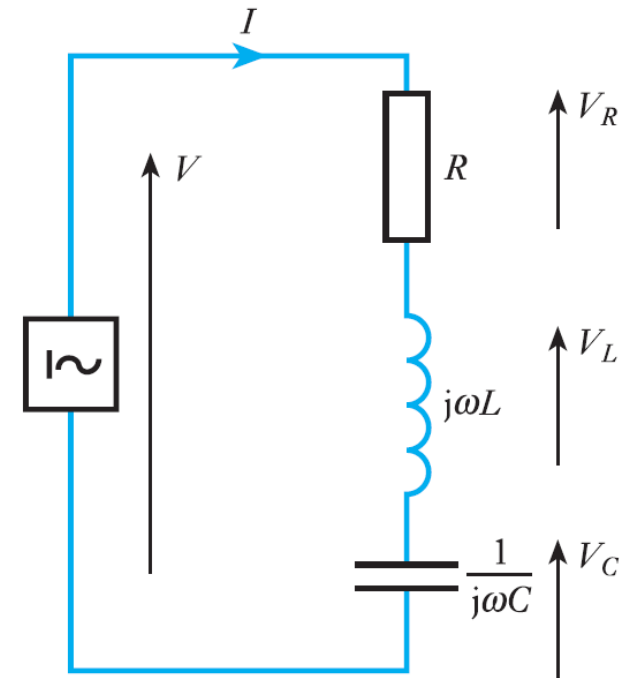
$$Q = \left( \frac{\text{Reactive power}}{\text{Average power}} \right)$$

- For inductive reactance  $X_L$  at resonance:

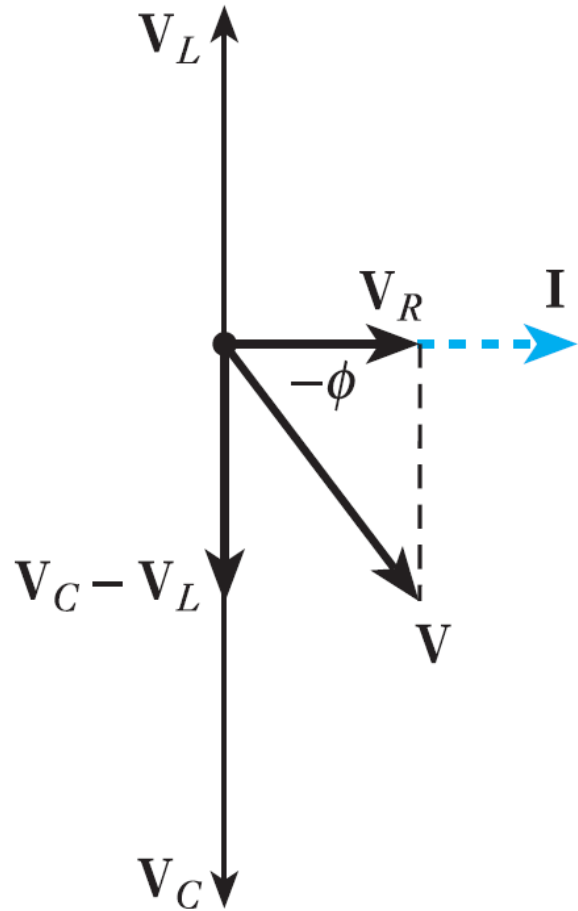
$$Q = \left( \frac{\text{Reactive power}}{\text{Average power}} \right) = \frac{I^2 X_L}{I^2 R} = \frac{\omega_r L}{R}$$

- For capacitive reactance  $X_C$  at resonance:

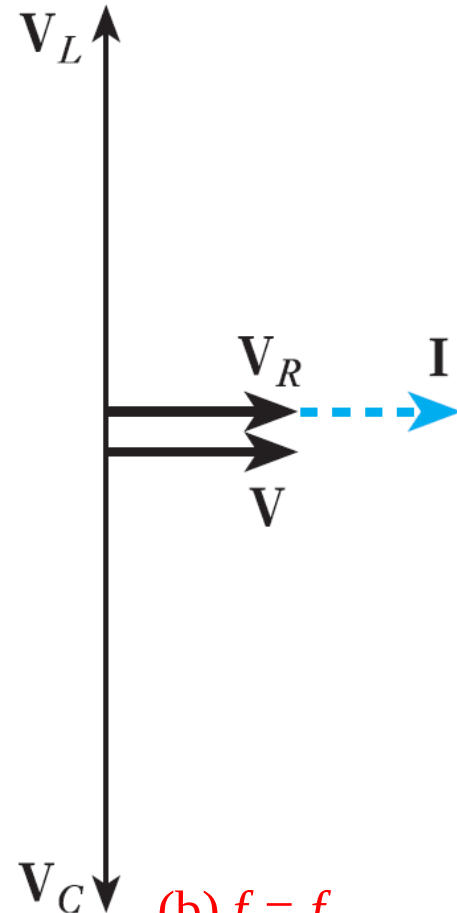
$$Q = \left( \frac{\text{Reactive power}}{\text{Average power}} \right) = \frac{I^2 X_C}{I^2 R} = \frac{1}{\omega_r C R}$$



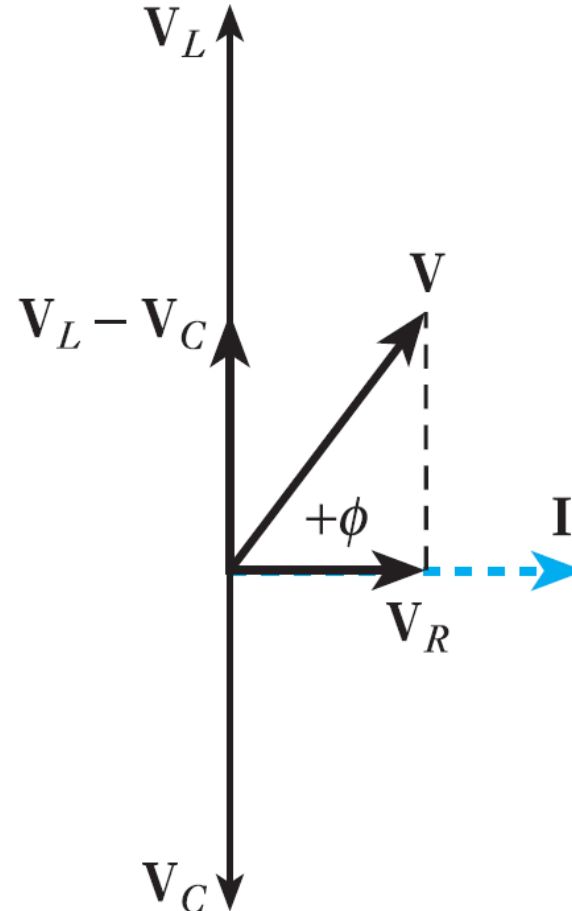
# VOLTAGES IN A SERIES *RLC* CIRCUIT



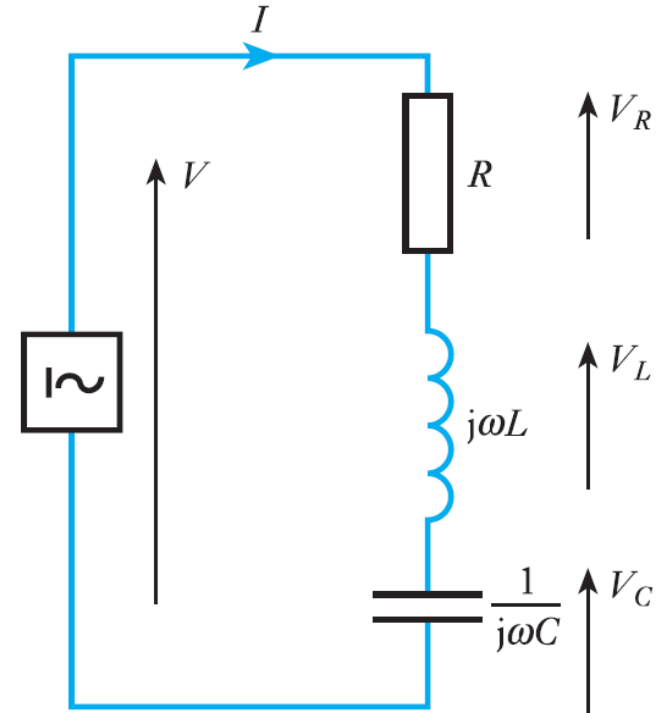
(a)  $f < f_r$   
Capacitive,  
 $I$  leads  $V$



(b)  $f = f_r$   
Resistive,  
 $V$  and  $I$  in  
phase



(c)  $f > f_r$   
Inductive,  
 $I$  lags  $V$



# VOLTAGES ACROSS *RLC* ELEMENTS AT RESONANCE

The voltage across resistor at  $f_r$  is,

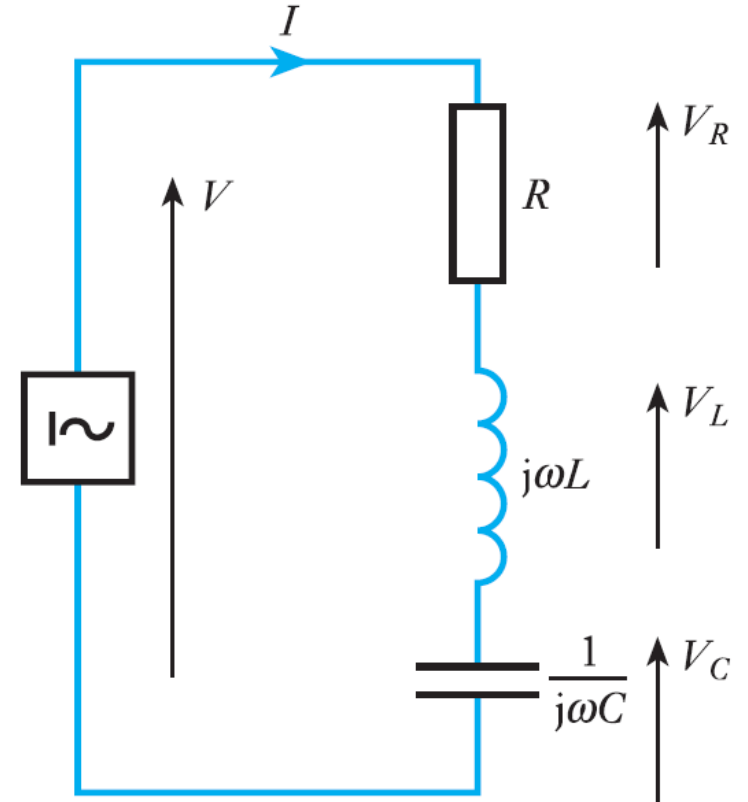
$$V_R = I_R \times R = I_m \times R = \frac{V}{R} \times R \Rightarrow V_R = V$$

The voltage across inductor at  $f_r$  is,

$$\begin{aligned} |V_L| &= X_L \times I_L = \omega_r L \times I_m = \omega_r L \times \frac{V}{R} = \frac{\omega_r L}{R} V = QV \\ \Rightarrow |V_L| &= QV \end{aligned}$$

The voltage across capacitor at  $f_r$  is,

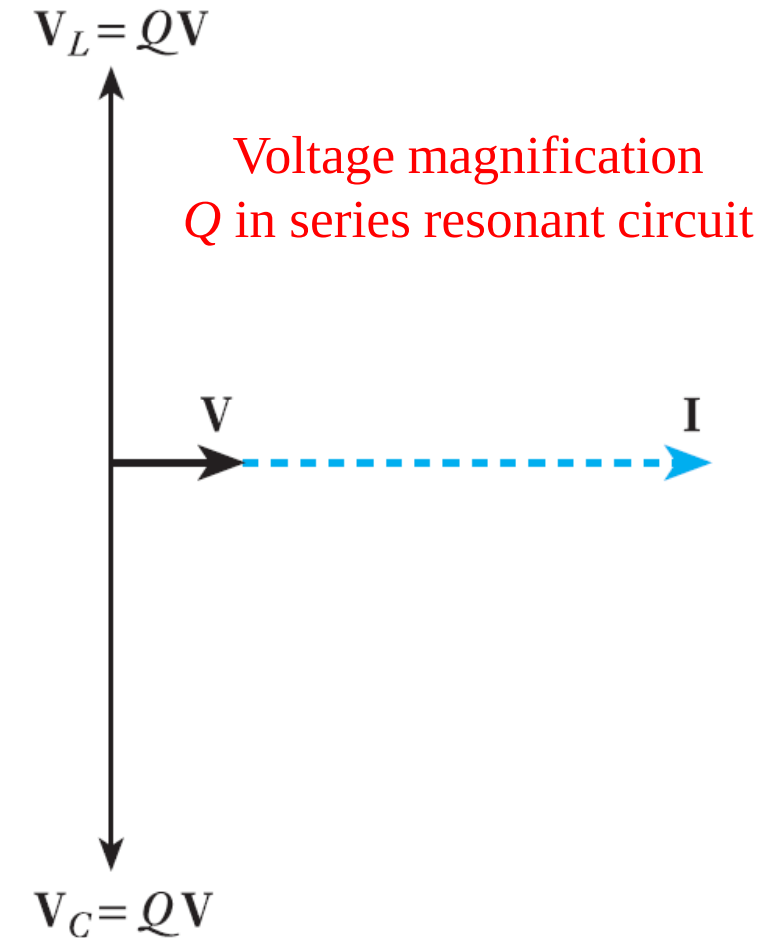
$$|V_C| = X_C \times I_C = \frac{1}{\omega_r C} \times I_m = \frac{1}{\omega_r C} \times \frac{V}{R} = \frac{1}{\omega_r CR} V = QV \Rightarrow |V_C| = QV$$



# VOLTAGES ACROSS *RLC* ELEMENTS AT RESONANCE

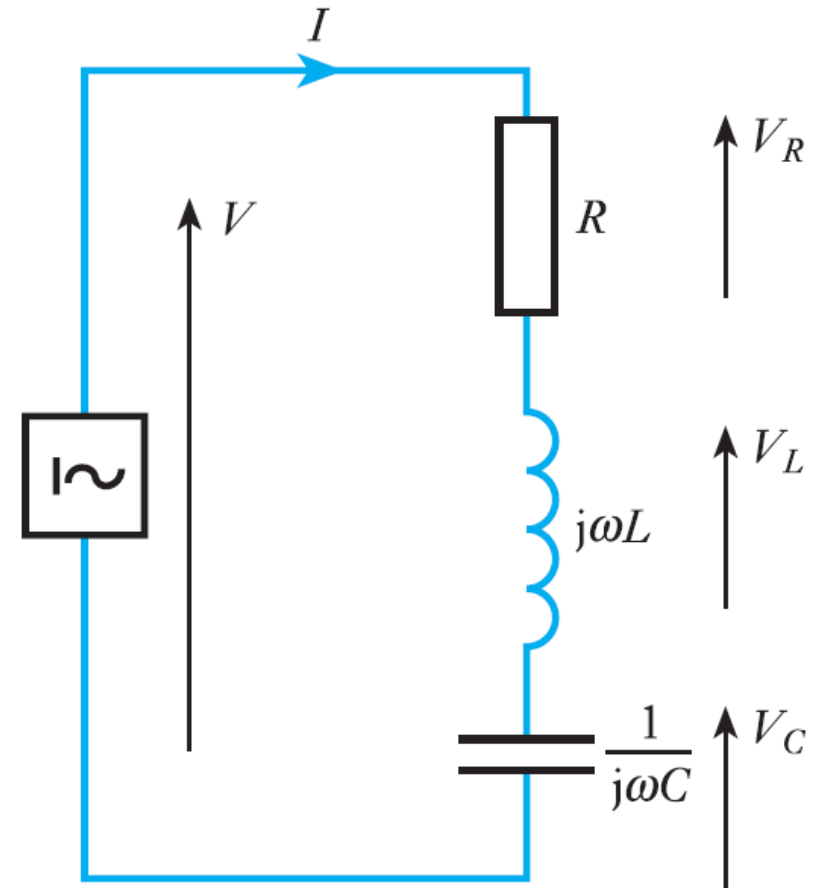
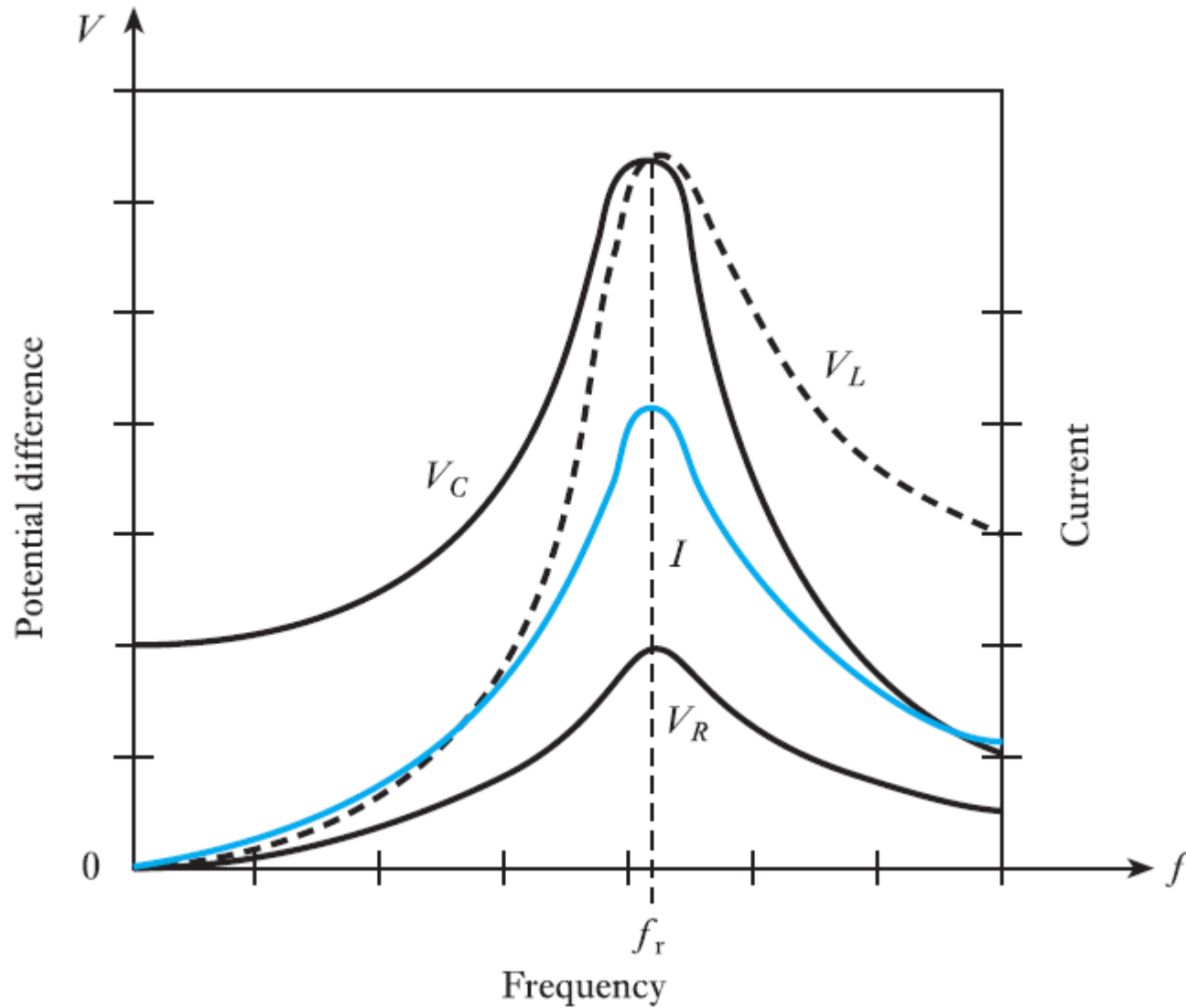
- $Q$  is termed as  $Q$  factor or voltage magnification, because  $V_C$  or  $V_L$  equals  $Q$  multiplied by the source voltage  $V$ .
- In a series *RLC* circuit, values of  $V_L$  and  $V_C$  can actually be very large at resonance and can lead to component damage if not recognized and subject to careful design.

$$Q = \frac{\omega_r L}{R} = \frac{1}{\omega_r C R} = \frac{1}{R} \sqrt{\left(\frac{L}{C}\right)}$$





# VOLTAGES ACROSS *RLC* ELEMENTS



Effect of frequency variation on voltages across  $R$ ,  $L$  and  $C$

# BANDWIDTH AND HALF POWER FREQUENCIES

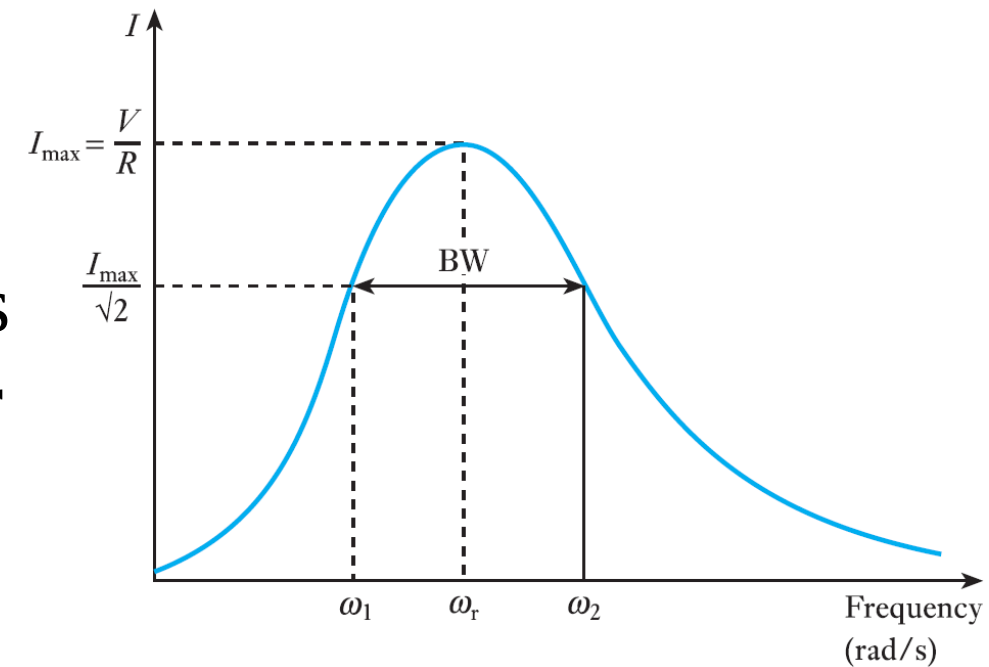
- In a series  $RLC$  circuit, at resonance, maximum power is drawn. i.e.,

$$P_r = I_{\max}^2 \times R; \text{ where } I_{\max} = \frac{V}{R} \text{ at resonance}$$

- Bandwidth represents the range of frequencies for which the power level in the signal is at least half of the maximum power.

$$\frac{P_r}{2} = \frac{I_{\max}^2 \times R}{2} = \left( \frac{I_{\max}}{\sqrt{2}} \right)^2 \times R$$

- The bandwidth of a circuit is also defined as the frequency range between the half-power points when  $I = I_{\max}/\sqrt{2}$ .



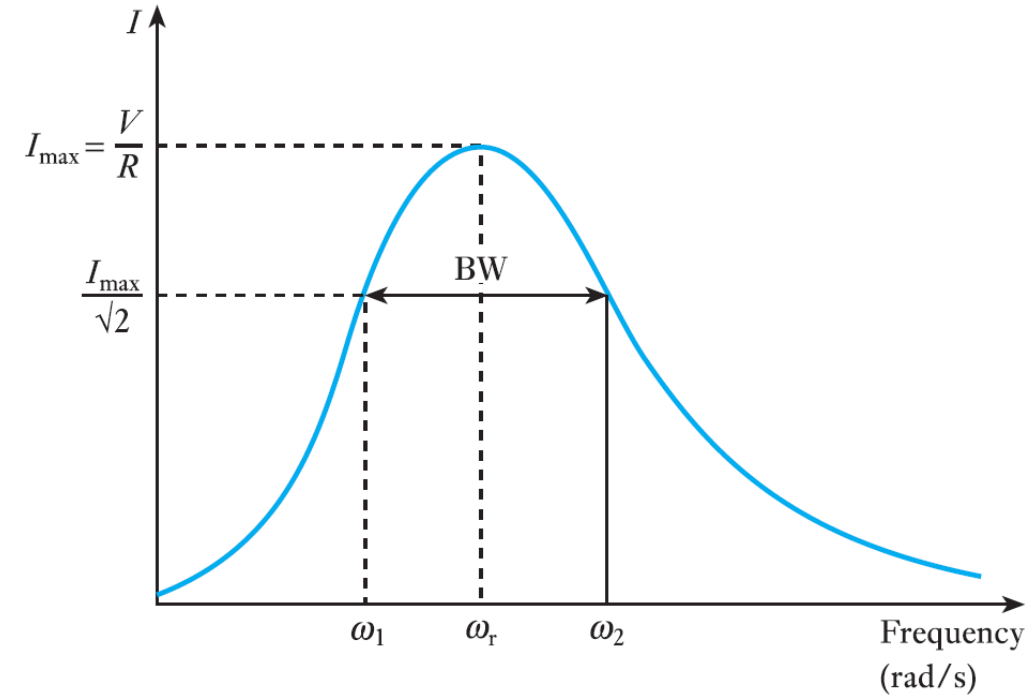
# BANDWIDTH AND HALF POWER FREQUENCIES

- Thus, the condition for half-power is given when

$$|I| = \frac{I_{\max}}{\sqrt{2}} = \frac{V}{R\sqrt{2}}$$

- The vertical lines either side of  $|I|$  indicate that only the magnitude of the current is under consideration – but the phase angle will not be neglected.
- The impedance corresponding to half power-points including phase angle is

$$Z(\omega_{1,2}) = R\sqrt{2} \angle \pm 45^\circ$$



The resonance peak, bandwidth and half-power frequencies

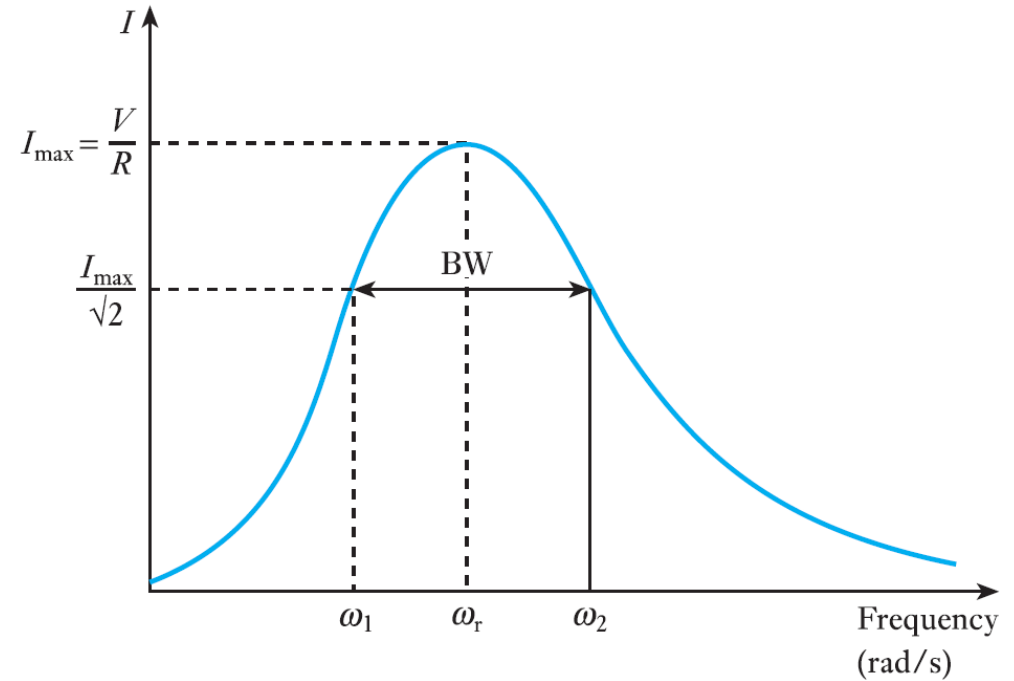
# BANDWIDTH AND HALF POWER FREQUENCIES

- The impedance in the complex form

$$Z(\omega_{1,2}) = R(1 \pm j1)$$

- Thus for half power,

$$I = \frac{V}{R(1 \pm j1)} \quad \text{and} \quad Z = R(1 \pm j1)$$



- At the half-power points, the phase angle of the current is  $45^\circ$ . Below the resonant frequency, at  $\omega_1$ , the circuit is capacitive and  $Z(\omega_1) = R(1 - j1)$ .
- Above the resonant frequency, at  $\omega_2$ , the circuit is inductive and  $Z(\omega_2) = R(1 + j1)$ .

# BANDWIDTH AND HALF POWER FREQUENCIES

- Now, the circuit impedance is given by,

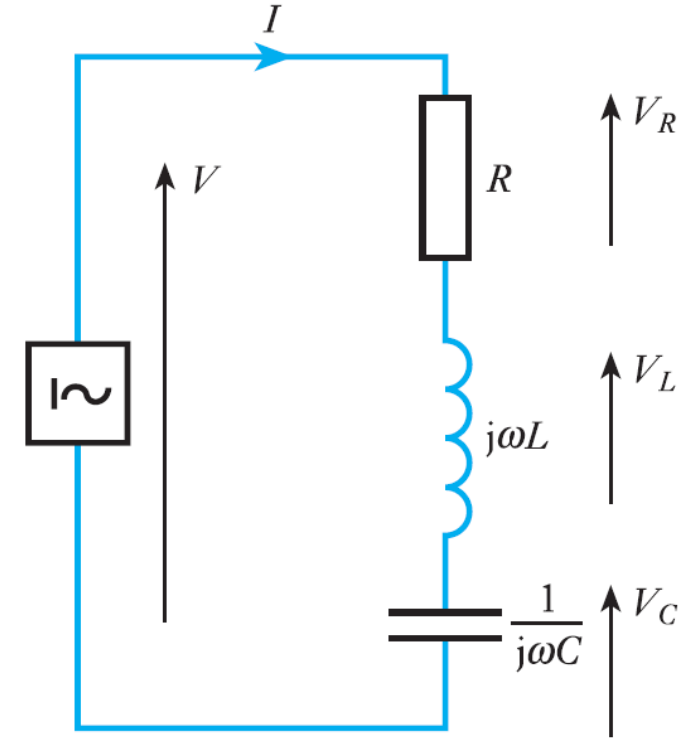
$$Z = R + j\left(\omega L - \frac{1}{\omega C}\right) = R\left(1 + j\left(\frac{\omega L}{R} - \frac{1}{\omega CR}\right)\right)$$

- At half power points,  $Z = R(1 \pm j1)$
- By comparison of above two equations, resulting in

$$\frac{\omega L}{R} - \frac{1}{\omega CR} = \pm 1$$

- As we know,

$$Q = \frac{\omega_r L}{R} = \frac{1}{\omega_r CR}$$



# BANDWIDTH AND HALF POWER FREQUENCIES

- Now, by multiplying and dividing with  $\omega_r$ :

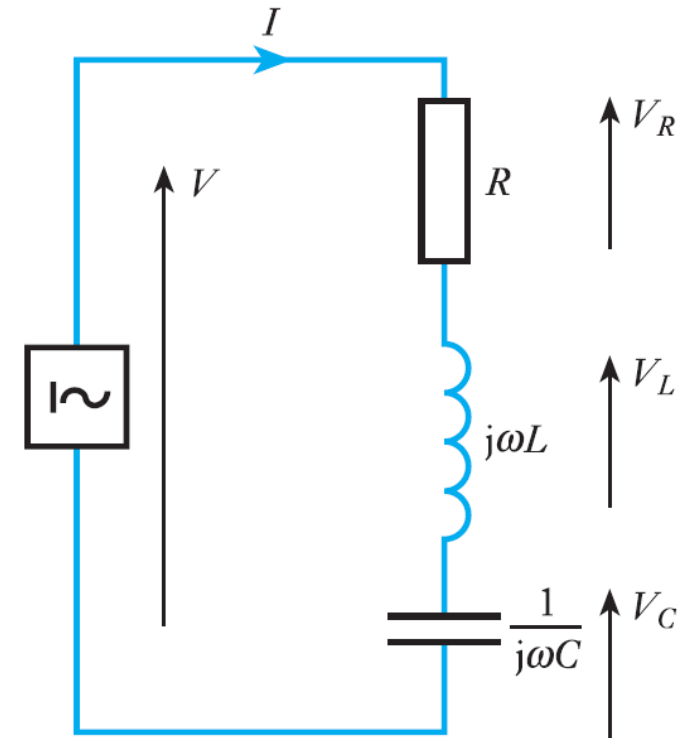
$$\frac{\omega L}{R} \frac{\omega_r}{\omega_r} - \frac{1}{\omega C R} \frac{\omega_r}{\omega_r} = \pm 1 \Rightarrow \frac{\omega}{\omega_r} Q - \frac{\omega_r}{\omega} Q = \pm 1 \Rightarrow Q \left( \frac{\omega}{\omega_r} - \frac{\omega_r}{\omega} \right) = \pm 1$$

- For  $\omega_2$ :

$$Q \left( \frac{\omega_2}{\omega_r} - \frac{\omega_r}{\omega_2} \right) = 1$$

- For  $\omega_1$ :

$$Q \left( \frac{\omega_1}{\omega_r} - \frac{\omega_r}{\omega_1} \right) = -1$$

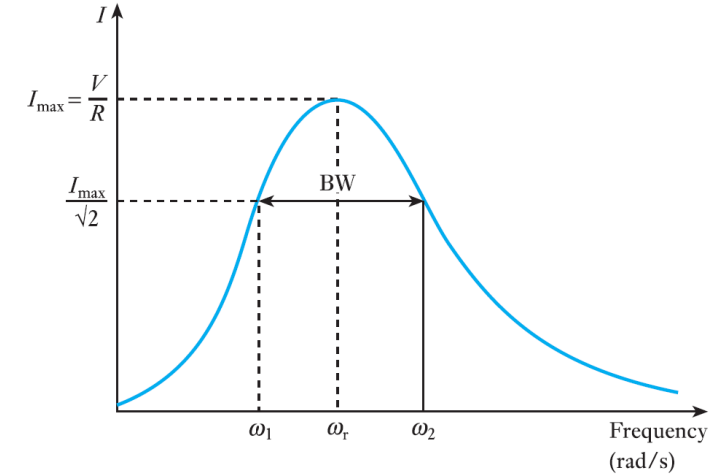


# BANDWIDTH AND HALF POWER FREQUENCIES

- The **half-power frequencies**  $\omega_2$  and  $\omega_1$  are obtained as,

$$\omega_2 = \frac{\omega_r}{2Q} + \omega_r \sqrt{1 + \frac{1}{4Q^2}}$$

$$\omega_1 = \frac{-\omega_r}{2Q} + \omega_r \sqrt{1 + \frac{1}{4Q^2}}$$

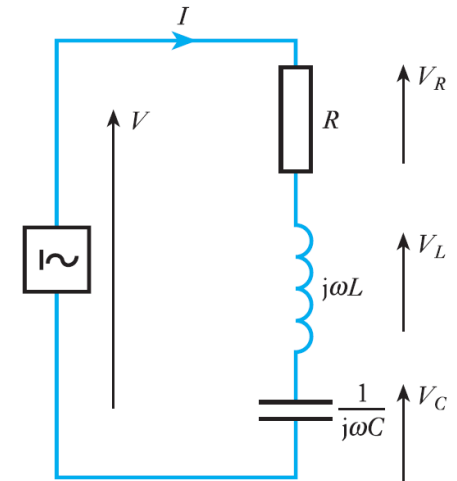


- The bandwidth is obtained as:

$$BW = \omega_2 - \omega_1 = \frac{\omega_r}{Q} \text{ i.e. Bandwidth} = \frac{\text{Resonant frequency}}{Q \text{ factor}}$$

- Resonant frequency in terms of  $\omega_2$  and  $\omega_1$ , is expressed as:

$$\omega_r = \sqrt{\omega_1 \omega_2}$$



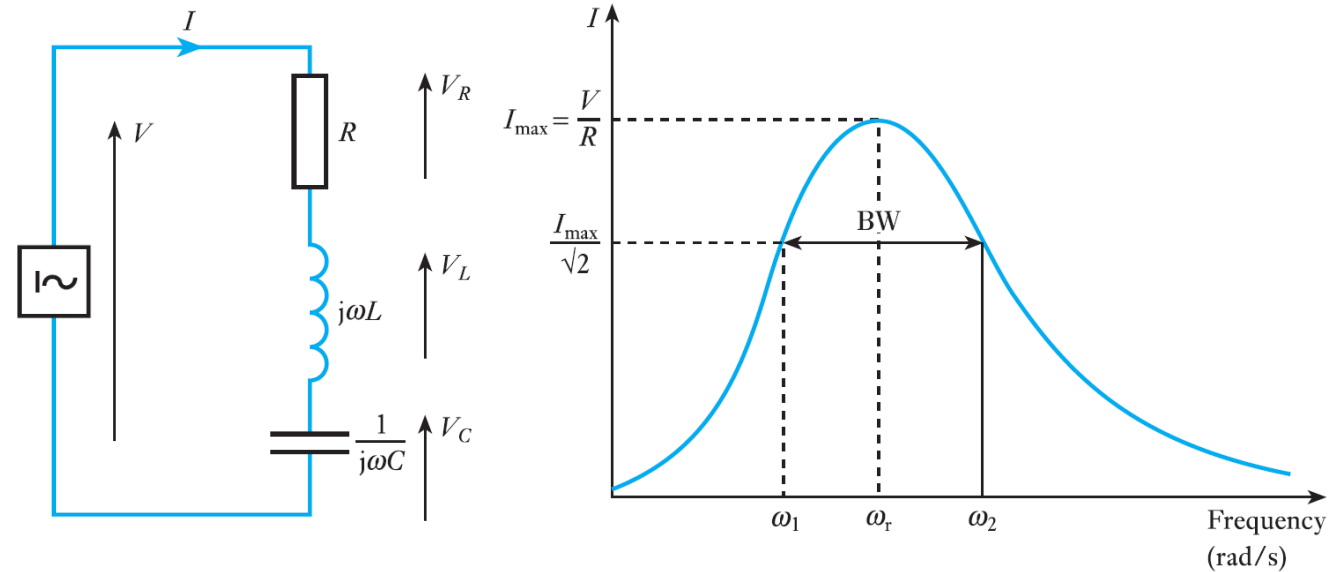
# BANDWIDTH AND HALF POWER FREQUENCIES

The bandwidth is also expressed as:

$$\omega_2 - \omega_1 = \frac{\omega_r}{Q} \Rightarrow \omega_2 - \omega_1 = \frac{R}{L} \text{ rad/s}$$

(or)

$$f_2 - f_1 = \frac{R}{2\pi L} \text{ Hz}$$



For  $Q \gg 1$ ,

$$\omega_r - \omega_1 \approx \frac{BW}{2} \Rightarrow \omega_1 = \omega_r - \frac{BW}{2} \Rightarrow \omega_1 = \omega_r - \frac{R}{2L} \text{ rad/s}$$

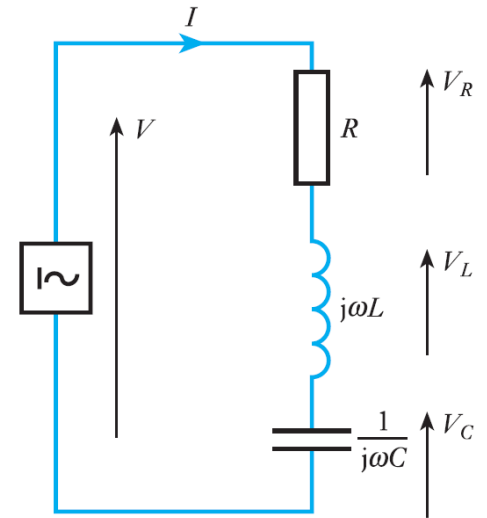
$$\omega_2 - \omega_r \approx \frac{BW}{2} \Rightarrow \omega_2 = \omega_r + \frac{BW}{2} \Rightarrow \omega_2 = \omega_r + \frac{R}{2L} \text{ rad/s}$$



# CONCLUSIONS

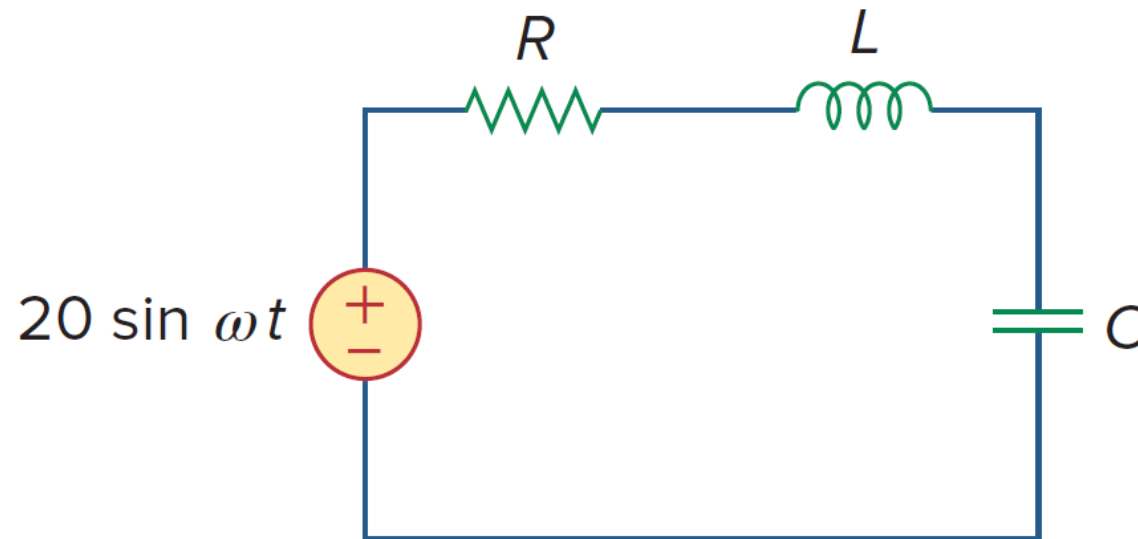
## Resonance in series $RLC$ circuit:

- The voltages which appear across the reactive components can be many times greater than that of the supply. The **factor of magnification**, the **voltage magnification** in the series circuit, is called the  **$Q$  factor**.
- An  $RLC$  series circuit accepts **maximum current** from the source **at resonance** and for that reason is called an **acceptor circuit**.



# PROBLEMS ON SERIES *RLC* CIRCUIT

- Q1.** In the circuit below,  $R = 2\ \Omega$ ,  $L = 1\ \text{mH}$ , and  $C = 0.4\ \mu\text{F}$ .
- (a) Find the resonant frequency  $\omega_r$  and the half-power frequencies  $\omega_1$  and  $\omega_2$ .
  - (b) Calculate the quality factor and bandwidth.
  - (c) Determine the amplitude of the current at  $\omega_r$ ,  $\omega_1$  and  $\omega_2$ .



## PROBLEMS ON SERIES *RLC* CIRCUIT

Ans:

(a) The resonant frequency is  $\omega_r = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{10^{-3} \times 0.4 \times 10^{-6}}} = 50 \text{ krad/s}$

The lower half-power frequency is

$$\omega_1 = -\frac{R}{2L} + \sqrt{\left(\frac{R}{2L}\right)^2 + \frac{1}{LC}} = -\frac{2}{2 \times 10^{-3}} + \sqrt{(10^3)^2 + (50 \times 10^3)^2} = 49 \text{ krad/s}$$

Similarly, the upper half-power frequency is

$$\omega_2 = \frac{R}{2L} + \sqrt{\left(\frac{R}{2L}\right)^2 + \frac{1}{LC}} = \frac{2}{2 \times 10^{-3}} + \sqrt{(10^3)^2 + (50 \times 10^3)^2} = 51 \text{ krad/s}$$

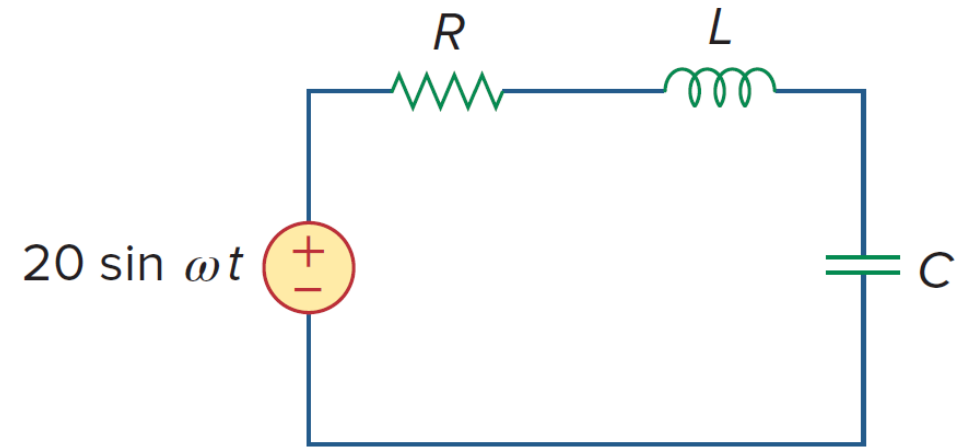
# PROBLEMS ON SERIES *RLC* CIRCUIT

(b) The bandwidth is  $BW = \omega_2 - \omega_1 = 2 \text{ krad/s}$

The quality factor is  $Q = \frac{\omega_r}{BW} = \frac{50}{2} = 25$

(c) At  $\omega = \omega_r$ :  $I = \frac{V_m}{R} = \frac{20}{2} = 10 \text{ A}$

At  $\omega = \omega_1, \omega_2$ :  $I = \frac{V_m}{\sqrt{2}R} = \frac{20}{\sqrt{2} \times 2} = 7.071 \text{ A}$



## PROBLEMS ON SERIES *RLC* CIRCUIT

**Q2.** A circuit, having a resistance of  $4.0\ \Omega$  with an inductance of  $0.5\ \text{H}$  and a variable capacitance in series, is connected across a  $100\ \text{V}$ ,  $50\ \text{Hz}$  supply. Calculate:

- (a) the capacitance required to attain resonance;
- (b) voltages across the inductance and the capacitance at resonance;
- (c) the  $Q$  factor of the circuit.

# PROBLEMS ON SERIES *RLC* CIRCUIT

Answer:

$$(a) \text{ For resonance: } 2\pi f_r L = \frac{1}{2\pi f_r C} \Rightarrow C = \frac{1}{(2\pi \times 50)^2 \times 0.5} = 20.3 \mu F$$

$$(b) \text{ At resonance: } I = \frac{V}{R} = \frac{100}{4} = 25 \text{ A}$$

$$\text{Voltage across inductance, } V_L = 2\pi \times 50 \times 0.5 \times 25 = 3927 \text{ V}$$

$$V_C = IX_C = \frac{25}{2\pi \times 50 \times 20.3 \times (10)^{-6}} \approx 3927 \text{ V}$$

$$(c) Q = \frac{X_L}{R} = \frac{2\pi \times 50 \times 0.5}{4} = 39.26$$

## PROBLEMS ON SERIES *RLC* CIRCUIT

**Q3.** The bandwidth of a series resonant circuit is 500 Hz. If the resonant frequency is 6000 Hz, what is the  $Q$ -factor? If  $R = 10\ \Omega$ , what is the value of the inductive reactance at resonance? Calculate the inductance and capacitance of the circuit.

# PROBLEMS ON SERIES *RLC* CIRCUIT

Answer:

$$\text{Bandwidth} = \frac{\text{resonant frequency}}{Q \text{ factor}}$$

$$\text{Hence, } Q = \frac{f_r}{\text{BW}} = \frac{6000}{500} = 12$$

$$Q = \frac{X_L}{R} \Rightarrow X_L = Q \times R = 12 \times 10 = 120 \, \Omega$$

$$\text{and } X_L = 2\pi f_r L \Rightarrow L = \frac{X_L}{2\pi f_r} = \frac{120}{2\pi \times 6000} = 3.18 \, \text{mH}$$

$$\text{and } |X_L| = |X_C| = 120 \, \Omega$$

$$X_C = \frac{1}{2\pi f_r C} = 120 \, \Omega$$

$$C = \frac{1}{2\pi \times 6000 \times 120} = 0.22 \, \mu\text{F}$$



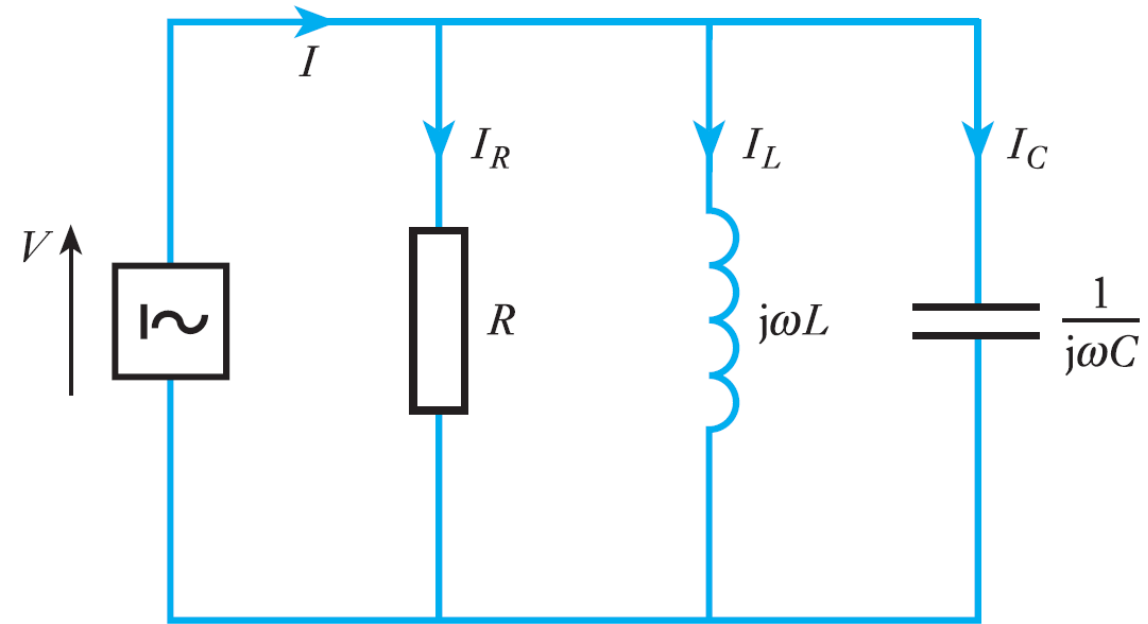
# RESONANCE IN PARALLEL *RLC* CIRCUIT

- The supply voltage:  $V = IZ$  where  $Z$  is the net impedance of the three parallel branches.
- In parallel circuits, it is simpler to consider the total admittance  $Y$  of the three branches. Thus,

$$V = IZ = \frac{I}{Y}$$

where

$$Y = G + \frac{1}{j\omega L} + j\omega C = G - \frac{j}{\omega L} + j\omega C = G + j\left(\omega C - \frac{1}{\omega L}\right)$$



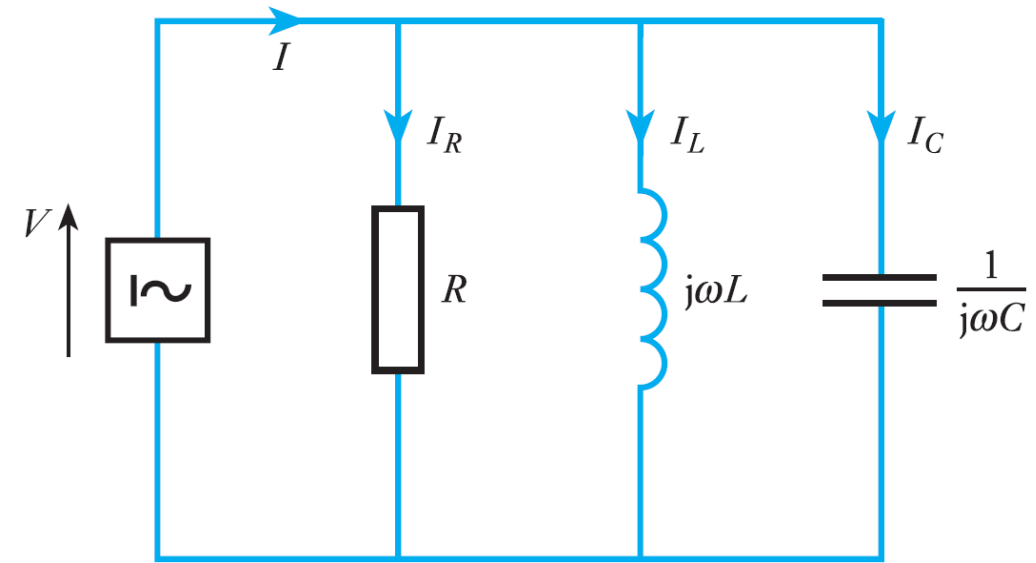
# RESONANCE IN PARALLEL *RLC* CIRCUIT

- At resonance ( $\omega = \omega_r$ ), the net susceptance is zero.

i.e. 
$$\left( \omega C - \frac{1}{\omega L} \right) = 0$$

- Therefore, the resonant frequency ( $\omega_r$ ) :

$$\omega_r = \frac{1}{\sqrt{LC}} \text{ rad/s}$$



- At the resonant frequency,  $Y = G = 1/R$ , the conductance of the parallel resistance, and  $I = VG$ .

# CURRENT THROUGH RESISTANCE

- The supply voltage magnitude:

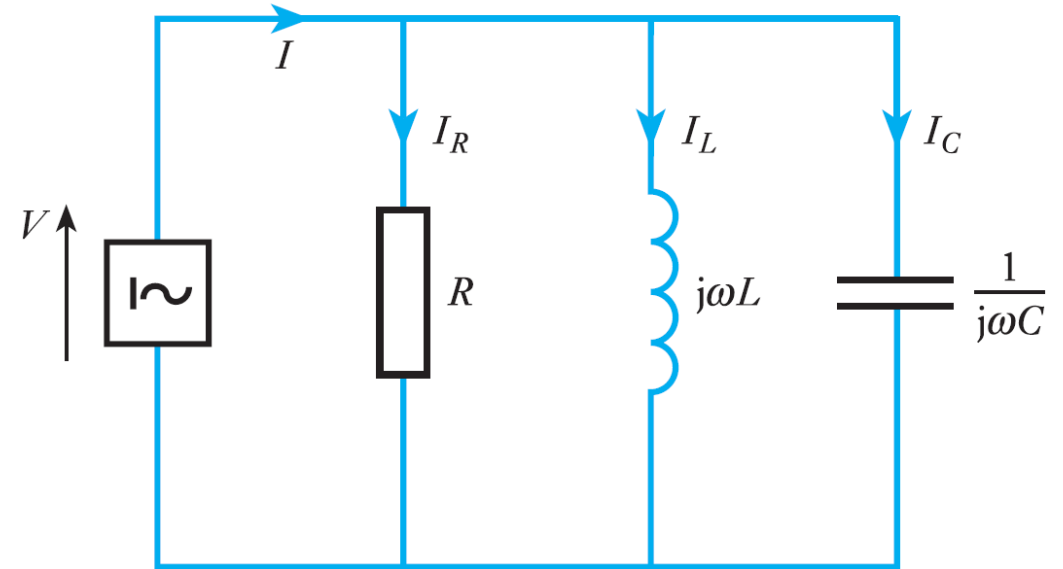
$$|V| = \frac{|I|}{\sqrt{\frac{1}{R^2} + \left(\omega C - \frac{1}{\omega L}\right)^2}}$$

- At resonance,  $\omega = \omega_r$ ,

$$|V| = \frac{|I|}{\sqrt{\frac{1}{R^2} + (0)^2}} \Rightarrow |V| = |I| \times R$$

- Current through the resistance at  $\omega_r$ :

$$I_R = \frac{V_R}{R} = \frac{V}{R} = \frac{I \times R}{R} \Rightarrow I_R = I$$



The three-branch  
parallel resonant circuit

# CURRENT MAGNIFICATION

- Magnitude of current through inductor at  $\omega_r$ :

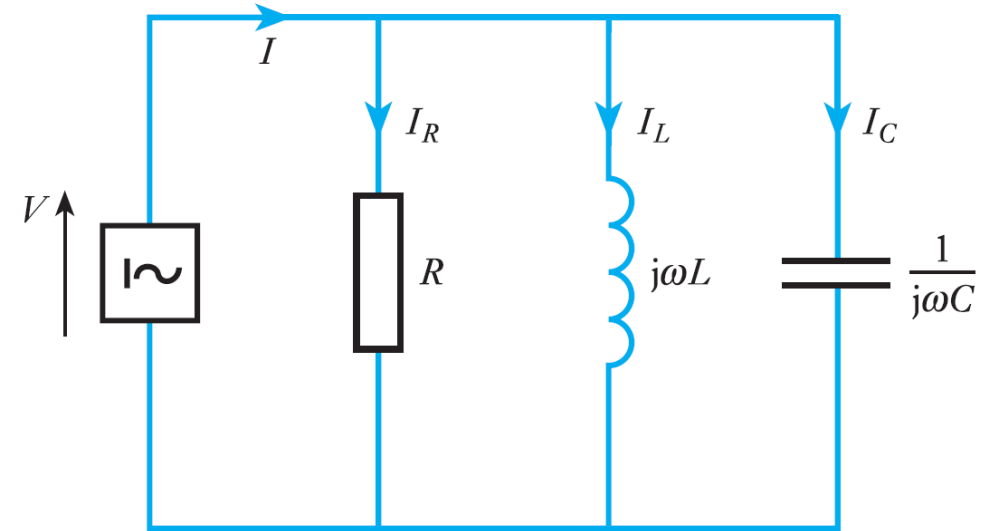
$$|I_L| = \frac{V}{X_L} = \frac{I \times R}{\omega_r L} = \left( \frac{R}{\omega_r L} \right) \times I = Q \times I$$

- Magnitude of current through capacitor at  $\omega_r$ :

$$|I_C| = \frac{V}{X_C} = \frac{I \times R}{\frac{1}{\omega_r C}} = (\omega_r CR) \times I = Q \times I$$

where  $Q$  is the current magnification i.e.,

$$Q = \left( \frac{R}{\omega_r L} \right) = (\omega_r CR)$$



The three-branch parallel resonant circuit

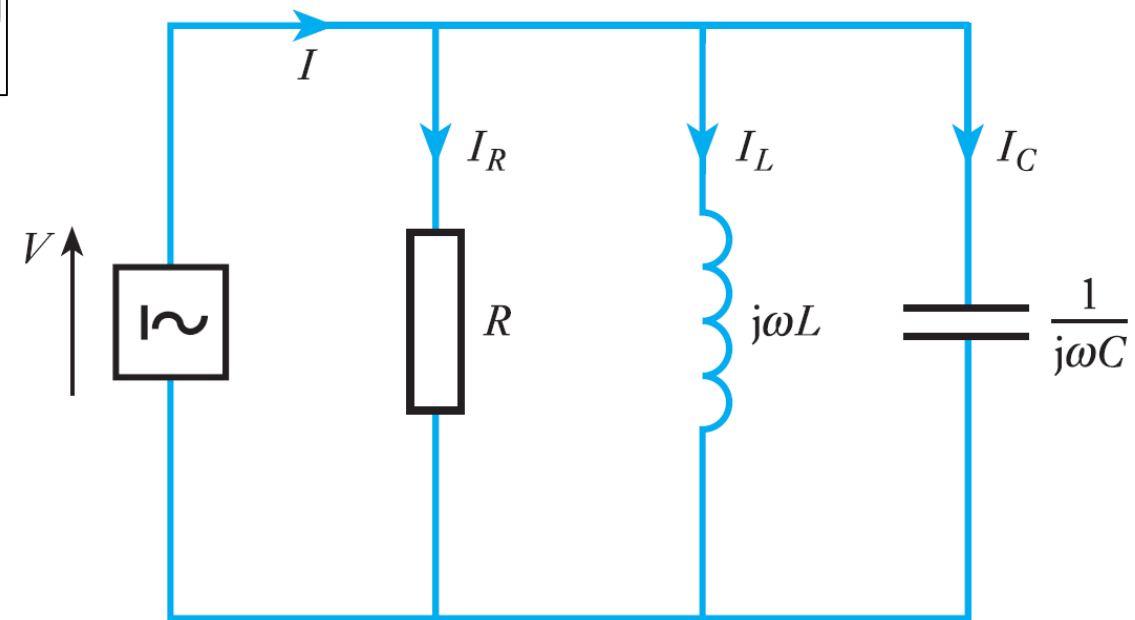
# CURRENT MAGNIFICATION

Current magnification  $Q$  is also expressed in terms of inductive or capacitive susceptance ( $B$ ), inductive or capacitive reactance ( $X$ ) and conductance ( $G$ ) :

$$Q = \left( \frac{1}{\omega_r L G} \right) = \left( \frac{\omega_r C}{G} \right) = \left( \frac{B}{G} \right) = \left( \frac{R}{X} \right)$$

By substituting  $\omega_r = 1/\sqrt{LC}$  in  $Q$  :

$$Q = \frac{1}{G} \sqrt{\frac{C}{L}} = R \sqrt{\frac{C}{L}}$$



The three-branch parallel resonant circuit

# BANDWIDTH AND HALF POWER FREQUENCIES

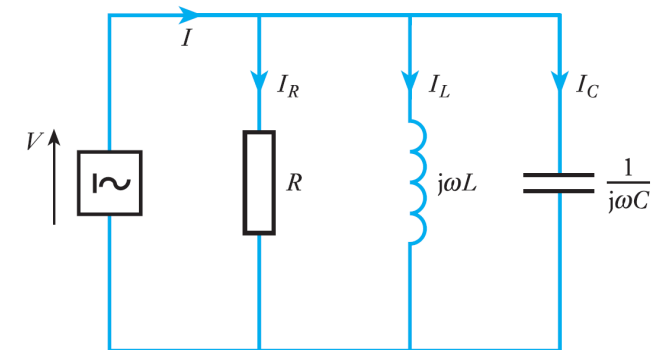
The parallel  $RLC$  circuit is the dual of the series  $RLC$  circuit. Therefore, by replacing  $R$ ,  $L$ , and  $C$  in the expressions for the series circuit with  $1/R$ ,  $C$ , and  $L$  respectively, we obtain for the parallel circuit, the  $Y_{\min}/2^{1/2}$  frequencies:

$$\omega_1 = \frac{-1}{2RC} + \sqrt{\left(\frac{1}{2RC}\right)^2 + \frac{1}{LC}}$$

$$\omega_2 = \frac{1}{2RC} + \sqrt{\left(\frac{1}{2RC}\right)^2 + \frac{1}{LC}}$$

- **Bandwidth:**  $BW = \omega_2 - \omega_1 = \frac{1}{RC}$

- Relation between BW and  $Q$ :  $Q = \frac{\omega_r}{BW} = \omega_r RC = \frac{R}{\omega_r L}$



# BANDWIDTH AND HALF POWER FREQUENCIES

The half-power frequencies in terms of quality factor:

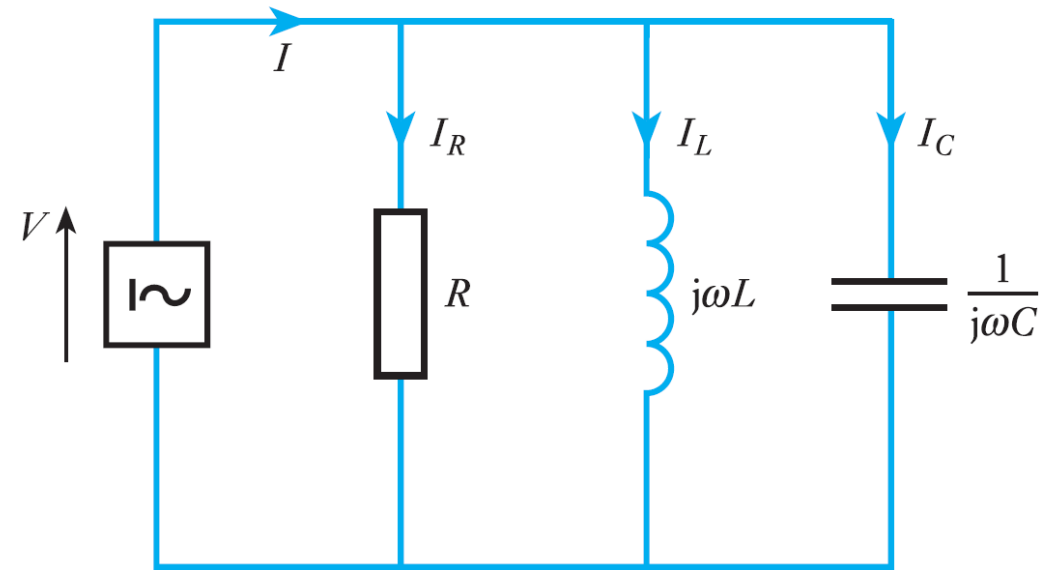
$$\omega_1 = \omega_r \sqrt{1 + \left(\frac{1}{2Q}\right)^2} - \frac{\omega_r}{2Q}$$

$$\omega_2 = \omega_r \sqrt{1 + \left(\frac{1}{2Q}\right)^2} + \frac{\omega_r}{2Q}$$

For  $Q \gg 1$ ,

$$\omega_r - \omega_1 = \frac{\text{BW}}{2}$$

$$\omega_2 - \omega_r = \frac{\text{BW}}{2}$$



The three-branch parallel resonant circuit

# PRACTICAL PARALLEL RESONANCE CIRCUIT

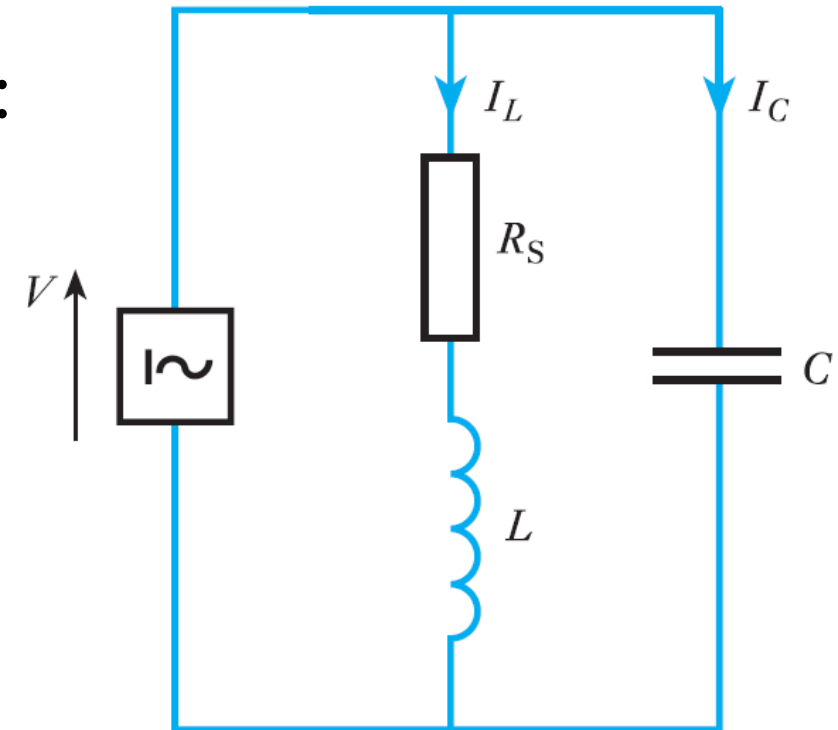
## Computation of resonant frequency of a “tank circuit”:

- The Figure shown is the two branch parallel resonant circuit. Also called **tank circuit**.
- The total admittance ( $Y_T$ ) of the circuit shown is:

$$Y_T = Y_1 + Y_2 = \left( \frac{1}{R_S + jX_L} \right) + \left( \frac{1}{-jX_C} \right),$$

$$Y_T = \left( \frac{R_S - jX_L}{R_S^2 + X_L^2} \right) + \left( \frac{j}{X_C} \right),$$

$$Y_T = \left( \frac{R_S}{R_S^2 + X_L^2} \right) + j \left( \frac{1}{X_C} - \frac{X_L}{R_S^2 + X_L^2} \right),$$



The two-branch  
parallel resonant circuit  
(or) tank circuit



# PRACTICAL PARALLEL RESONANCE CIRCUIT

Computation of resonant frequency of a “tank circuit”:

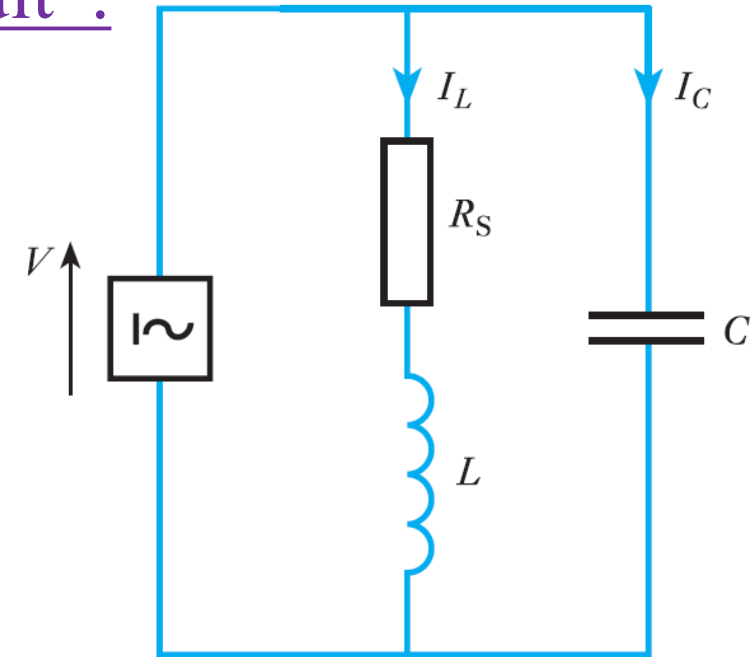
$$Y_T = \left( \frac{R_S}{R_S^2 + X_L^2} \right) + j \left( \frac{1}{X_C} - \frac{X_L}{R_S^2 + X_L^2} \right),$$

At resonance ( $\omega = \omega_r$ ), the net susceptance is zero.

$$\text{i.e. } \frac{1}{X_C} - \frac{X_L}{R_S^2 + X_L^2} = 0 \Rightarrow R_S^2 + X_L^2 = \omega_r L \times \frac{1}{\omega_r C},$$

$$R_S^2 + X_L^2 = \frac{L}{C}$$

$$R_S^2 + \omega_r^2 L^2 = \frac{L}{C} \Rightarrow \omega_r = \sqrt{\frac{1}{LC} - \frac{R_S^2}{L^2}} \text{ rad/s}$$



The two-branch  
parallel resonant circuit  
(or) tank circuit

# PRACTICAL PARALLEL RESONANCE CIRCUIT

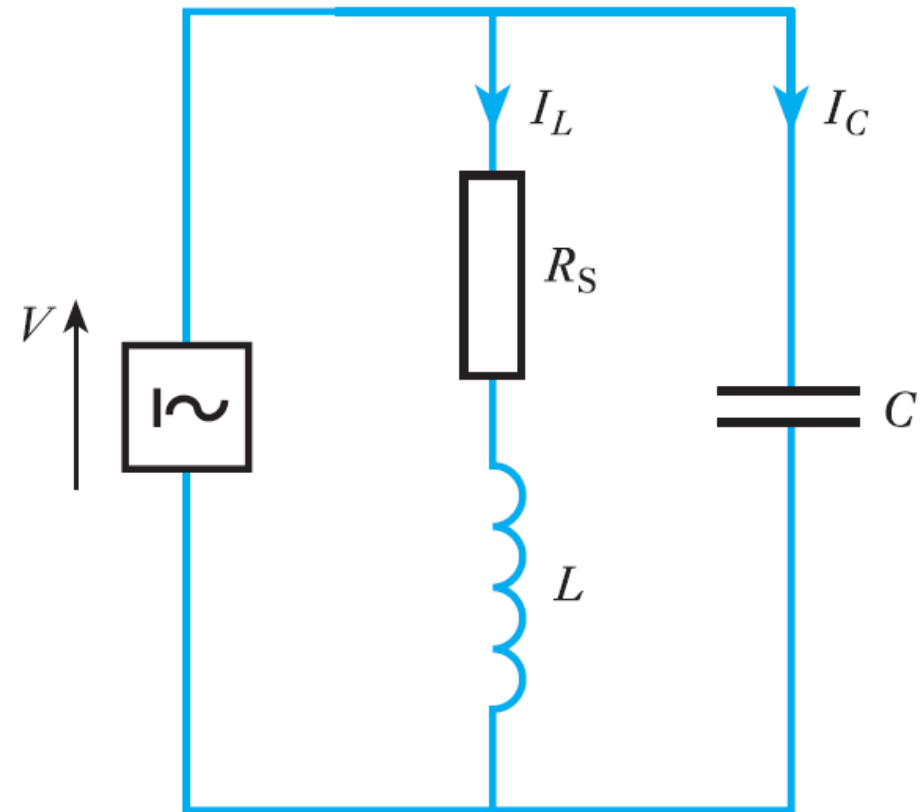
## Computation of resonant frequency of a “tank circuit”:

- The resonant frequency in Hz is:

$$f_r = \frac{1}{2\pi} \sqrt{\frac{1}{LC} - \frac{R_S^2}{L^2}} \text{ Hz}$$

- The admittance at resonance is:

$$Y_T(f = f_r) = \left( \frac{R_S}{R_S^2 + X_L^2} \right) = \frac{R_S C}{L},$$



The two-branch parallel resonant circuit (or) tank circuit

# IDEAL TANK CIRCUIT

## Computation of resonant frequency of a “ideal tank circuit”:

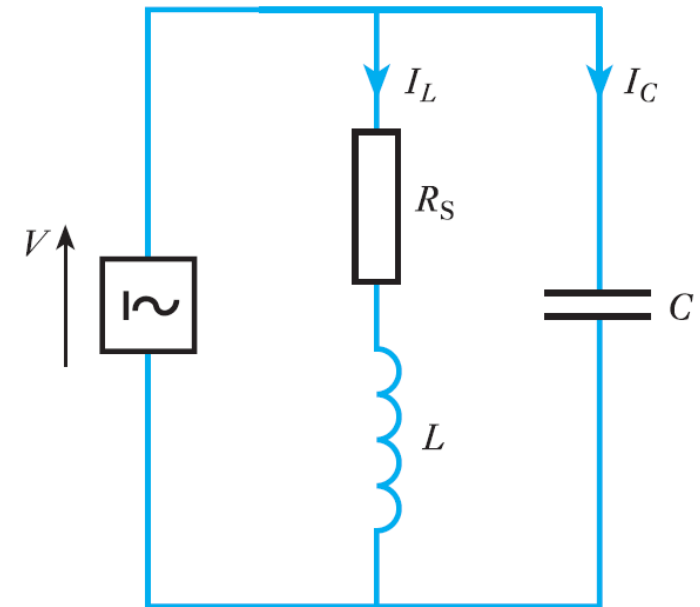
- In ideal tank circuit, the series resistance  $R_S$  is made zero.
- The total admittance ( $Y_T$ ) of the circuit shown ( $R_S = 0$ ) is:

$$Y_T = Y_L + Y_C = \left( \frac{1}{jX_L} \right) + \left( \frac{1}{-jX_C} \right),$$

$$Y_T = +j \left( \frac{1}{X_C} - \frac{1}{X_L} \right).$$

- At resonance ( $\omega = \omega_r$ ), the net susceptance is zero.

$$\text{i.e. } \frac{1}{X_C} - \frac{1}{X_L} = 0 \Rightarrow \omega_r = \frac{1}{\sqrt{LC}} \text{ rad/s; } f_r = \frac{1}{2\pi\sqrt{LC}} \text{ Hz}$$



Ideal tank circuit  
with  $R_S = 0$

# CONCLUSIONS

## Resonance in parallel $RLC$ circuit:

- The lowest current from the source occurs at the resonant frequency of a parallel circuit hence it is called a **rejector circuit**.
- At resonance, **the current in the branches of the parallel circuit** can be **many times greater than the supply current**.
- The **factor of magnification**, the **current magnification** in the parallel circuit, is again called the  **$Q$  factor**.
- **At the resonant frequency** of a resonant parallel network, **the impedance is wholly resistive**. The value of this impedance is known as the **dynamic resistance or dynamic impedance**.

# SUMMARY OF THE CHARACTERISTICS OF RESONANT *RLC* CIRCUITS

| Characteristics                                          | Series circuit                                                            | Parallel circuit                                                          |
|----------------------------------------------------------|---------------------------------------------------------------------------|---------------------------------------------------------------------------|
| Resonant frequency, $\omega_r$                           | $\frac{1}{\sqrt{LC}}$                                                     | $\frac{1}{\sqrt{LC}}$                                                     |
| Quality factor, $Q$                                      | $\frac{\omega_r L}{R}$ or $\frac{1}{\omega_r CR}$                         | $\frac{R}{\omega_r L}$ or $\omega_r CR$                                   |
| Bandwidth, BW                                            | $\frac{\omega_r}{Q}$                                                      | $\frac{\omega_r}{Q}$                                                      |
| Half-power frequencies, $\omega_1, \omega_2$             | $\omega_r \sqrt{1 + \left(\frac{1}{2Q}\right)^2} \mp \frac{\omega_r}{2Q}$ | $\omega_r \sqrt{1 + \left(\frac{1}{2Q}\right)^2} \mp \frac{\omega_r}{2Q}$ |
| For high circuits, ( $Q \geq 10$ ), $\omega_1, \omega_2$ | $\omega_r \mp \frac{BW}{2}$                                               | $\omega_r \mp \frac{BW}{2}$                                               |

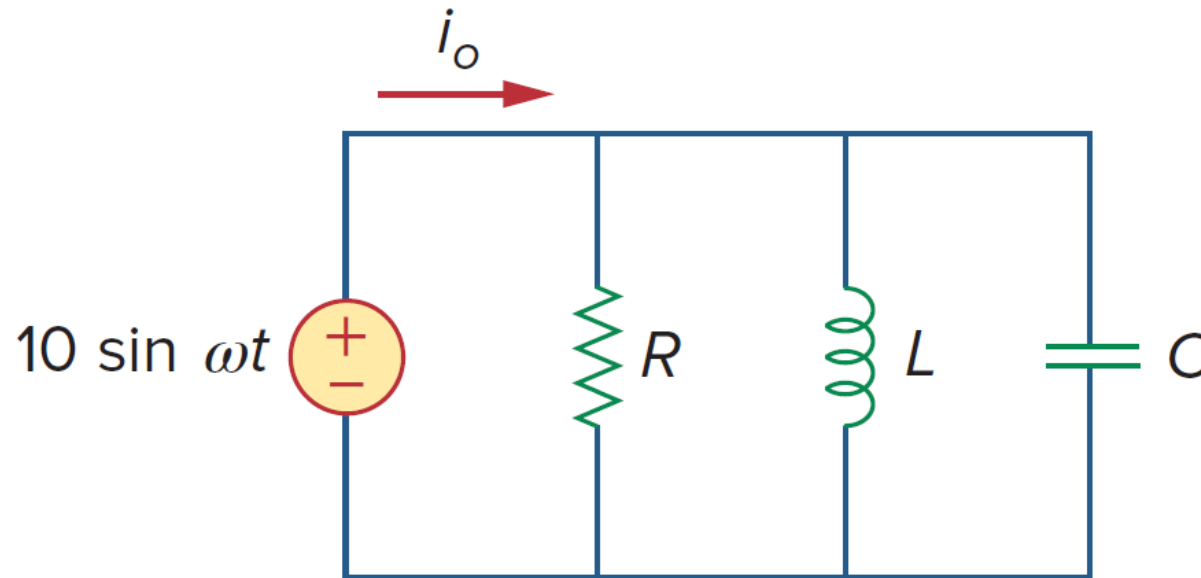
# PROBLEMS ON PARALLEL *RLC* CIRCUIT

**Q1.** In the parallel *RLC* circuit below,  $R = 8 \text{ k}\Omega$ ,  $L = 0.2 \text{ mH}$ , and  $C = 8 \text{ }\mu\text{F}$ .

(a) Calculate  $\omega_r$ ,  $Q$ , and BW.

(b) Find  $\omega_1$  and  $\omega_2$ .

(c) Determine the power dissipated at  $\omega_r$ ,  $\omega_1$ , and  $\omega_2$ .



# PROBLEMS ON PARALLEL *RLC* CIRCUIT

Answer:

$$\omega_r = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{0.2 \times 10^{-3} \times 8 \times 10^{-6}}} = \frac{10^5}{4} = 25 \text{ krad/s}$$

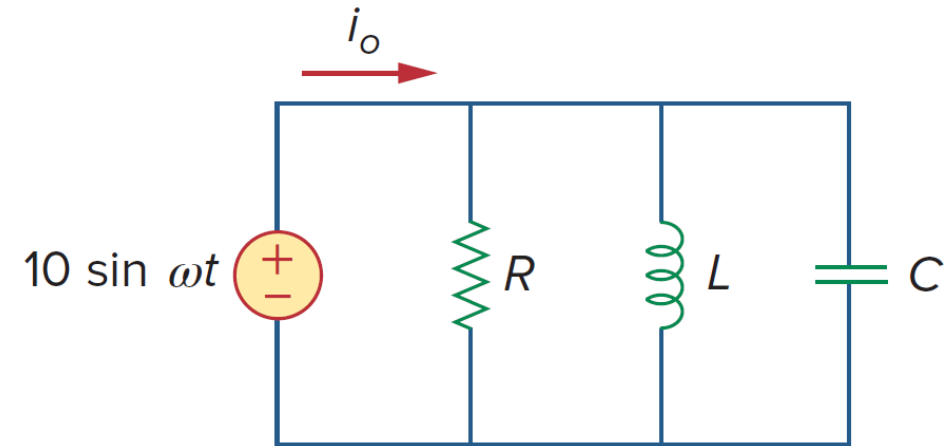
$$Q = \frac{R}{\omega_r L} = \frac{8 \times 10^3}{25 \times 10^3 \times 0.2 \times 10^{-3}} = 1600$$

$$BW = \frac{\omega_r}{Q} = 15.625 \text{ rad/s}$$

(b) Since  $Q \gg 1$ ,

$$\omega_1 = \omega_r - \frac{BW}{2} = 25000 - 7.812 = 24992 \text{ rad/s}$$

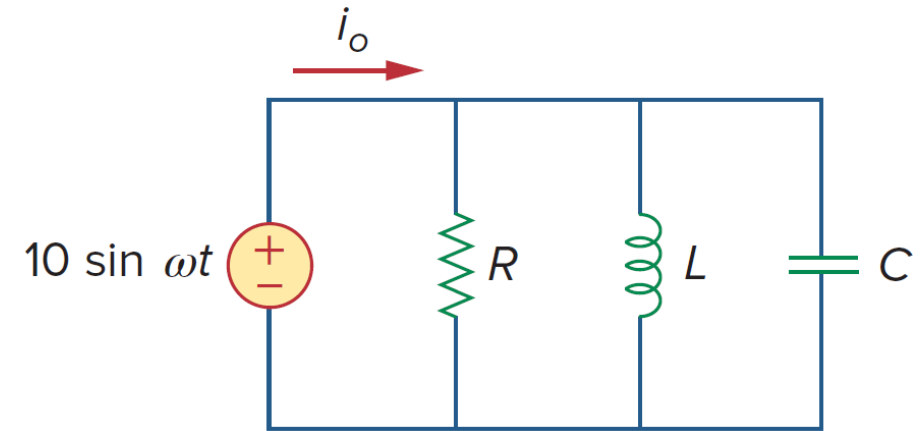
$$\omega_2 = \omega_r + \frac{BW}{2} = 25000 + 7.812 = 25008 \text{ rad/s}$$



## PROBLEMS ON PARALLEL *RLC* CIRCUIT

(c) At  $\omega = \omega_r$ ,  $Y = 1/R$  or  $Z = R = 8 \text{ k}\Omega$ . Then

$$I_0 = \frac{V}{Z} = \frac{10 \angle -90^\circ}{8000} = 1.25 \angle -90^\circ \text{ mA}$$



Since the entire current flows through  $R$  at resonance, the average power dissipated at  $\omega = \omega_r$  is

$$P = \frac{1}{2} |I_0|^2 R = \frac{1}{2} (1.25 \times 10^{-3})^2 (8 \times 10^3) = 6.25 \text{ mW}$$

or

$$P = \frac{V_m^2}{2R} = \frac{100}{2 \times 8 \times 10^3} = 6.25 \text{ mW}$$

At  $\omega = \omega_1, \omega_2$ ,

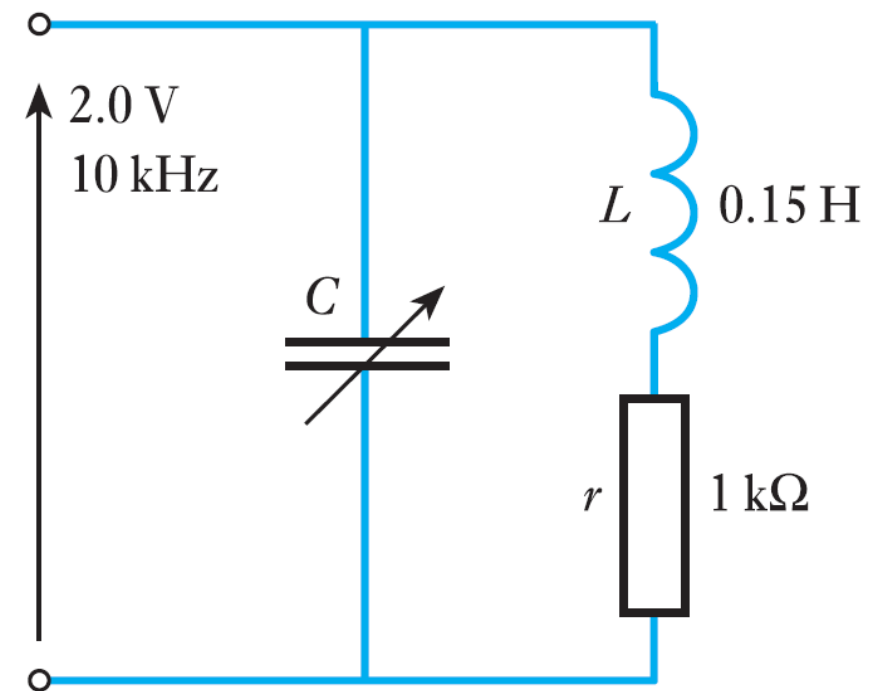
$$P = \frac{V_m^2}{2R} = 6.25 \text{ mW}$$



## PROBLEMS ON PARALLEL *RLC* CIRCUIT

**Q2.** A coil of  $1\text{ k}\Omega$  resistance and  $0.15\text{ H}$  inductance is connected in parallel with a variable capacitor across a  $2.0\text{ V}$ ,  $10\text{ kHz}$  a.c. supply as shown. Calculate:

- (a) the capacitance of the capacitor when the supply current is a minimum;
- (b) the effective impedance  $Z_r$  of the network at resonance;
- (c) the supply current at resonance.



# PROBLEMS ON PARALLEL *RLC* CIRCUIT

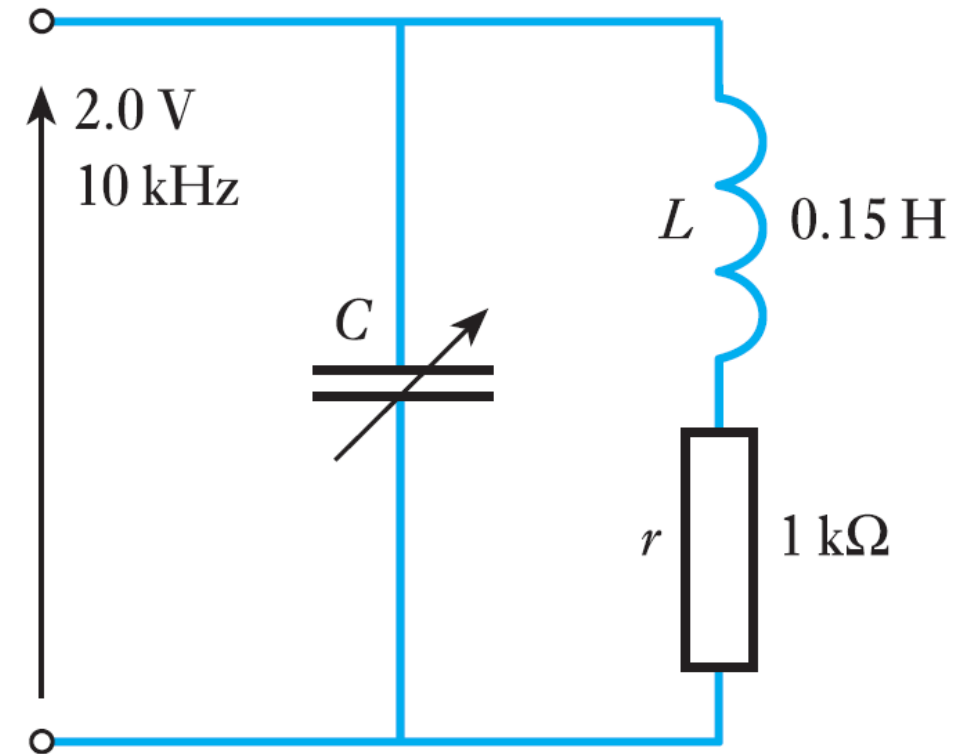
Ans:

$$(a) \quad f_r = \frac{1}{2\pi\sqrt{LC}} \Rightarrow f_r^2 = \frac{1}{4\pi^2 LC}$$

$$C = \frac{1}{4\pi^2 L f_r^2} = \frac{1}{4\pi^2 \times 0.15 \times 10^8} = 1.69 \text{ nF}$$

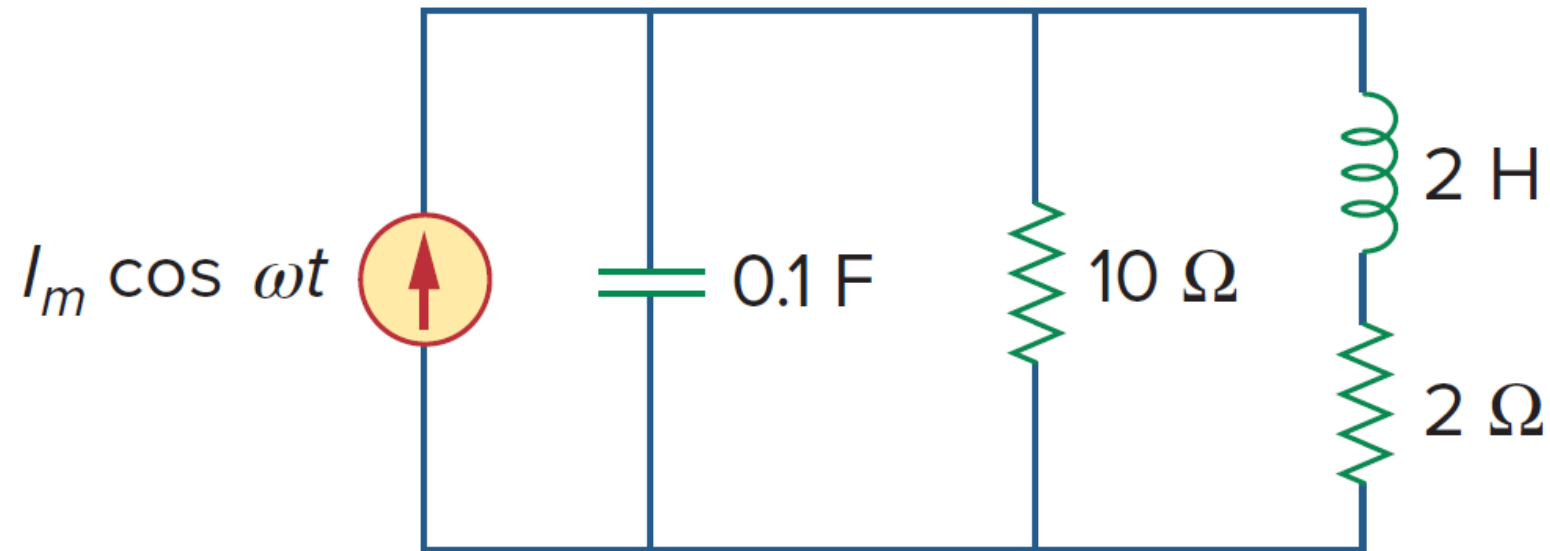
$$(b) \quad Z_r = \frac{L}{CR_s} = \frac{0.15}{1.69 \times 10^{-9} \times 1000} = 89 \text{ k}\Omega$$

$$(c) \quad I_s = \frac{V}{Z_r} = \frac{2}{89 \times 1000} = 22.5 \times 10^{-6} \text{ A}$$



# PROBLEMS ON PARALLEL *RLC* CIRCUIT

**Q3.** Determine the resonant frequency  $\omega_r$  of the circuit shown below.



# PROBLEMS ON PARALLEL *RLC* CIRCUIT

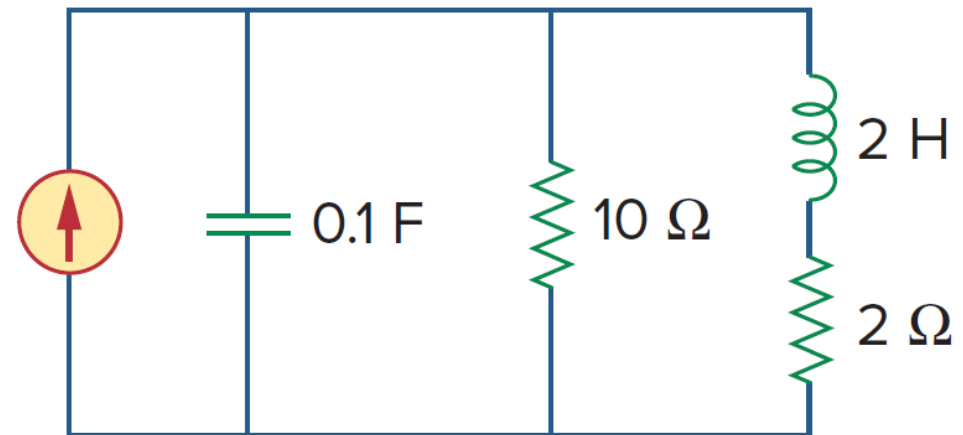
**Ans:** The input admittance is

$$Y = j\omega 0.1 + \frac{1}{10} + \frac{1}{2 + j\omega 2} = 0.1 + j\omega 0.1 + \frac{2 - j\omega 2}{4 + 4\omega^2}$$

$$Y = \left( 0.1 + \frac{2}{4 + 4\omega^2} \right) + j \left( 0.1\omega - \frac{2\omega}{4 + 4\omega^2} \right)$$

At resonance,  $\omega = \omega_r$ , the net susceptance is zero. i.e.,

$$0.1\omega_r - \frac{2\omega_r}{4 + 4\omega_r^2} = 0 \Rightarrow \omega_r = 2 \text{ rad/s.} \quad I_m \cos \omega t$$

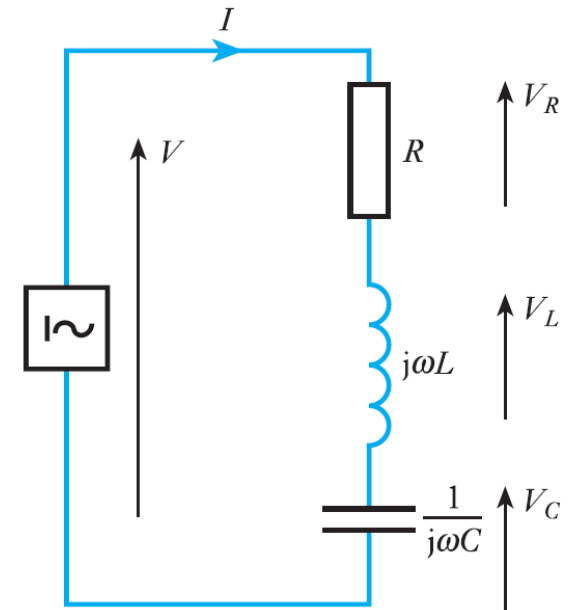


# DYNAMIC IMPEDANCE (OR) DYNAMIC RESISTANCE

- The dynamic impedance (dynamic resistance) is the resistance offered by the circuit to the input signal under resonance condition.

Q1. What is the dynamic impedance in a standard series RLC circuit?

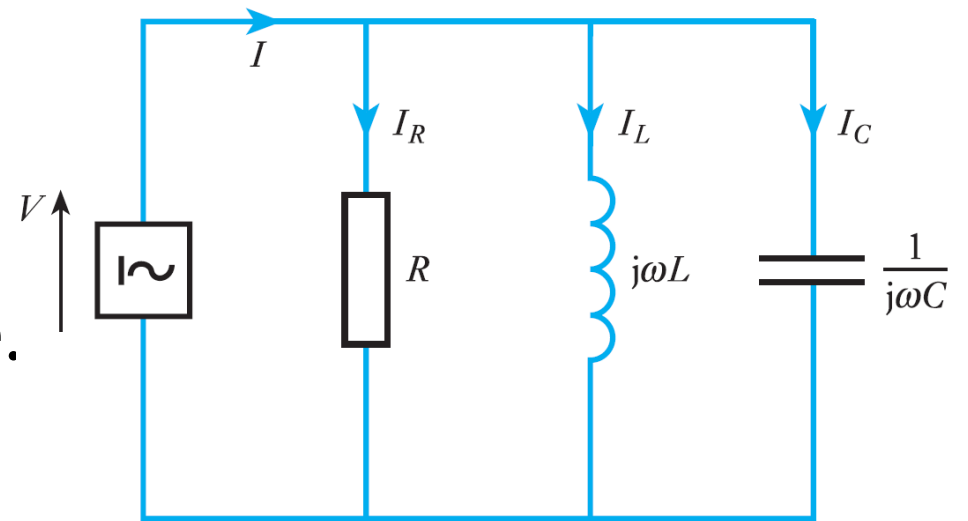
**Ans.** In a standard series RLC circuit, at resonance the net reactance becomes zero. Therefore, the input supply see only resistance. Hence, dynamic impedance  $Z_{dynamic} = R$ .



# DYNAMIC IMPEDANCE (OR) DYNAMIC RESISTANCE

Q2. What is the dynamic impedance in a general parallel RLC circuit?

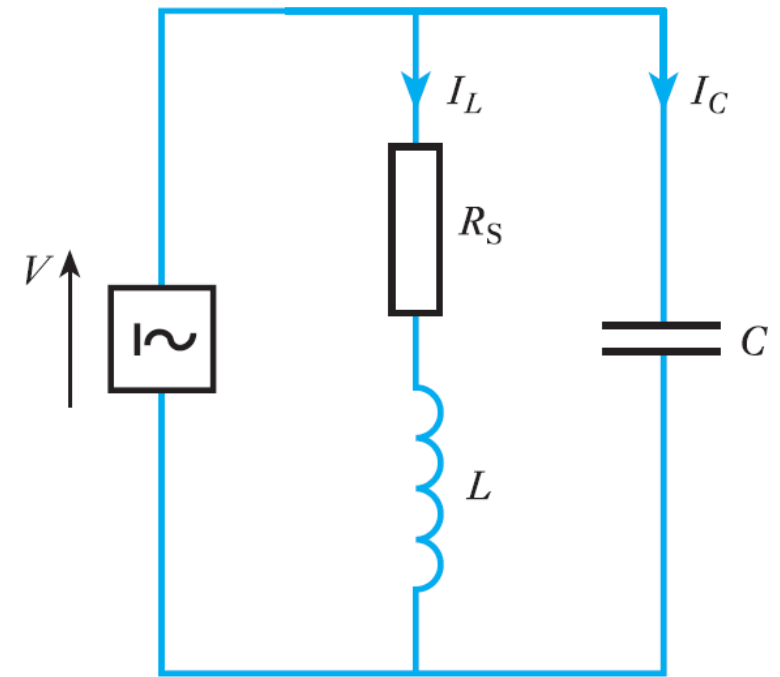
**Ans.** In a general parallel RLC circuit, at resonance, the net susceptance becomes zero. Therefore, the input supply see only resistance. Hence, dynamic impedance  $Z_{dynamic} = R$ .



# DYNAMIC IMPEDANCE (OR) DYNAMIC RESISTANCE

Q3. What is the dynamic impedance in a ideal tank circuit?

Ans. In ideal tank circuit, at resonance, the circuit acts like a open circuit. Because, in ideal tank circuit, the  $R_S = 0$ . Therefore, the dynamic impedance  $Z_{dynamic} \rightarrow \infty$



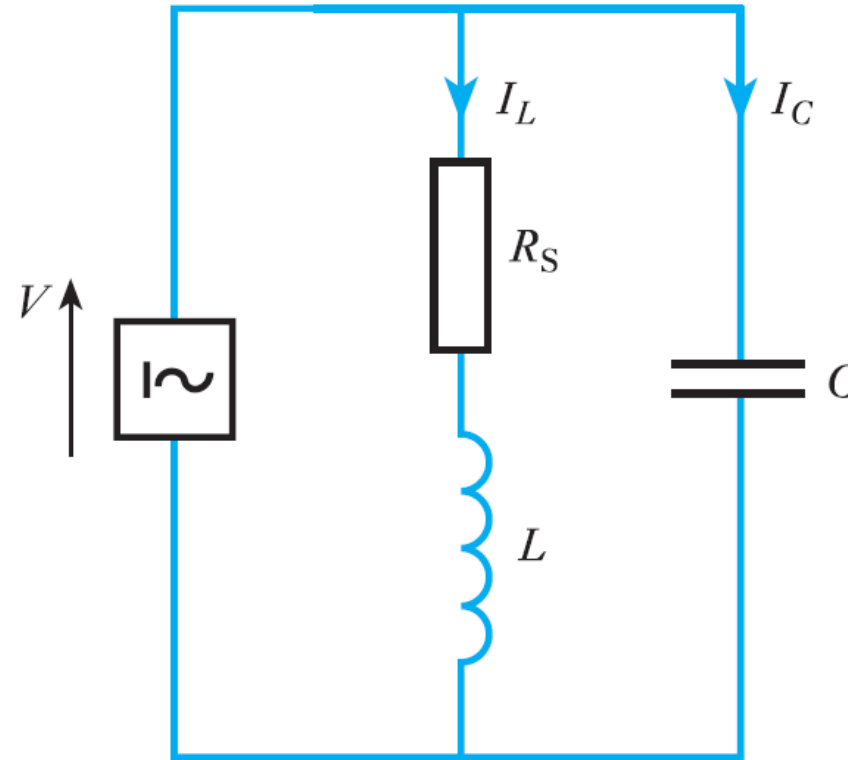
Ideal tank circuit with  $R_S = 0$

# DYNAMIC IMPEDANCE (OR) DYNAMIC RESISTANCE

**Q4.** Determine the dynamic impedance of a practical tank circuit?

**Ans.**

$$Z_{dynamic} = \frac{L}{R_S C}$$



Practical tank circuit



# MISCELLANEOUS PROBLEMS FOR PRACTICE

**Q1.** Two practical coils with internal resistances of  $R_1$ ,  $R_2$  has quality factors of  $Q_1$ ,  $Q_2$ , respectively. If these two coils are connected in series then resultant quality factor is.....

**Ans.**

$$\frac{Q_1 R_1 + Q_2 R_2}{R_1 + R_2}$$

## MISCELLANEOUS PROBLEMS FOR PRACTICE

**Q2.** A series circuit consists of a  $0.5\ \mu\text{F}$  capacitor, a coil of inductance  $0.32\ \text{H}$  and resistance  $40\ \Omega$  and a  $20\ \Omega$  non-inductive resistor. Calculate the value of the resonant frequency of the circuit. When the circuit is connected to a  $30\ \text{V}$  a.c. supply at this resonant frequency, determine: (a) the p.d. across each of the three components; (b) the current flowing in the circuit; (c) the active power absorbed by the circuit.

**Answer.**

$400\ \text{V}$ ,  $401\ \text{V}$ ,  $10\ \text{V}$ ,  $0.5\ \text{A}$ ,  $15\ \text{W}$ .

## MISCELLANEOUS PROBLEMS FOR PRACTICE

**Q3.** A circuit consists of a  $10\ \Omega$  resistor, a  $30\ \text{mH}$  inductor and a  $1\ \mu\text{F}$  capacitor, and is supplied from a  $10\ \text{V}$  variable-frequency source. Find the frequency for which the voltage developed across the capacitor is a maximum and calculate the magnitude of this voltage.

**Answer.**  $920\ \text{Hz}$ ,  $173\ \text{V}$ .

## MISCELLANEOUS PROBLEMS FOR PRACTICE

**Q4.** A series circuit comprises an inductor, of resistance  $10\ \Omega$  and inductance  $159\ \mu\text{H}$ , and a variable capacitor connected to a  $50\ \text{mV}$  sinusoidal supply of frequency  $1\ \text{MHz}$ . What value of capacitance will result in resonant conditions and what will then be the current? For what values of capacitance will the current at this frequency be reduced to 10 per cent of its value at resonance?

**Answer.**

$159\ \text{pF}$ ,  $5\ \text{mA}$ ;  $145\ \text{pF}$ ,  $177\ \text{pF}$ .

## MISCELLANEOUS PROBLEMS FOR PRACTICE

**Q5.** A coil, of resistance  $R$  and inductance  $L$ , is connected in series with a capacitor  $C$  across a variable-frequency source. The voltage is maintained constant at 300 mV and the frequency is varied until a maximum current of 5 mA flows through the circuit at 6 kHz. If, under these conditions, the  $Q$  factor of the circuit is 105, calculate: (a) the voltage across the capacitor; (b) the values of  $R$ ,  $L$  and  $C$ .

**Answer.**

31.5 V, 60  $\Omega$ , 0.167 H, 4220 pF.

# MISCELLANEOUS PROBLEMS FOR PRACTICE

**Q6.** Calculate the resonant frequency of the circuit in Fig.

**Answer.** 173.21 rad/s.

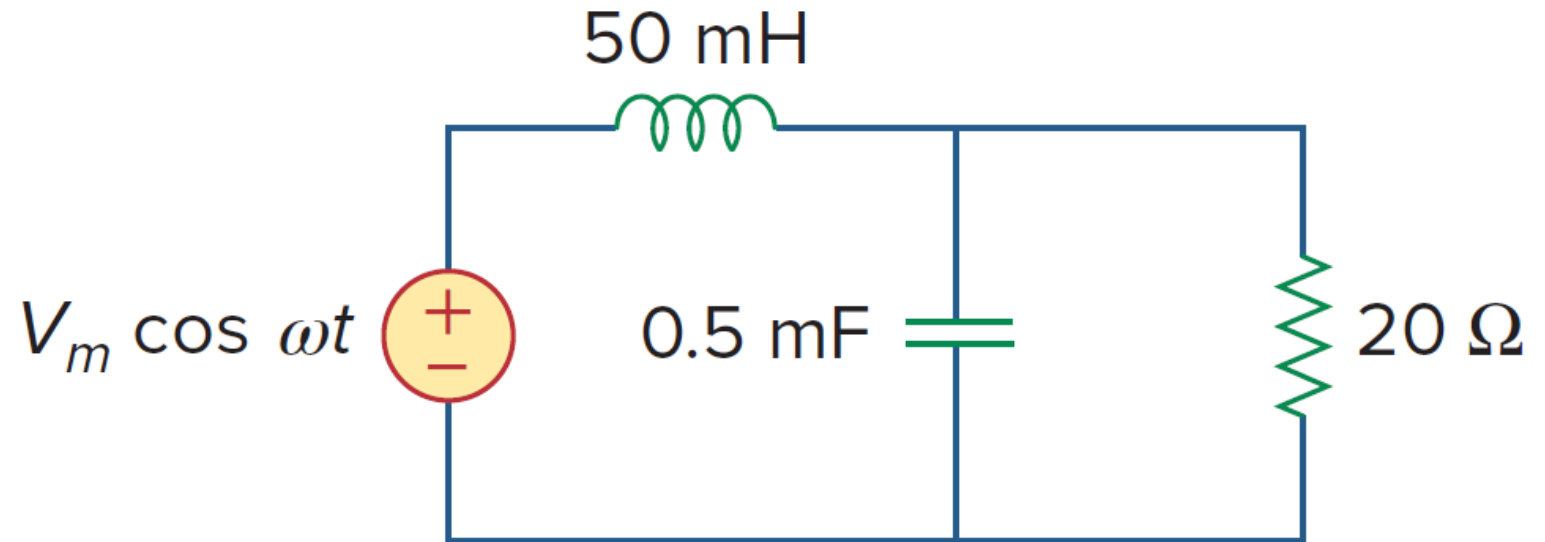


Fig. For Q6.

## PRACTICE PROBLEM ON SERIES *RLC* CIRCUIT

**Q7.** A series-connected circuit has  $R = 4 \, \Omega$  and  $L = 25 \, \text{mH}$ . (a) Calculate the value of  $C$  that will produce a quality factor of 50. (b) Find  $\omega_1$ ,  $\omega_2$ , and BW. (c) Determine the average power dissipated at  $\omega = \omega_r, \omega_1, \omega_2$ .

Take  $V_m = 100 \, \text{V}$ .

**Answer.**

(a)  $0.625 \, \mu\text{F}$ , (b)  $7920 \, \text{rad/s}$ ,  $8080 \, \text{rad/s}$ ,  $160 \, \text{rad/s}$ , (c)  $1.25 \, \text{kW}$ ,  $0.625 \, \text{kW}$ ,  $0.625 \, \text{kW}$ .

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